

Transparent layer constancy is improved by motion, stereo disparity, highly regular background pattern, and successive presentation

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The visual system uses figural and colorimetric regularities in the retinal image to recognize optical filters and to discern the properties of the transparent overlay from properties of the background. Previous work suggests that the perceived color and transmittance of the transparent layer vary less under illumination changes than it would be expected from corresponding changes in the input. Here, we tested how the degree of this approximate transparent layer constancy (TLC) depends on factors that presumably facilitate the decomposition into a filter and a background layer. Using an asymmetric filter matching task, we found that motion, stereo disparity, and a highly regular background pattern each contribute to the vividness of the transparency impression and the degree of TLC. Combining these cues led to a cumulative increase in TLC, suggesting a “strong fusion” cue integration process. We also tested objects with invalid figural conditions for transparency (T-junctions). The tendency to perceive these objects as opaque and to establish a proximal match increased the more conspicuous the violation of this figural condition was. Furthermore, we investigated the gain in TLC due to alternating presentation. Alternating presentation enhanced TLC and color constancy to a comparable degree, and our results suggest that adaptation contributes to this effect.

Introduction

In everyday life, we reliably recognize objects of interest despite changes in illumination, changed viewing angle, or partial occlusion. This performance is quite surprising, because such context variations usually lead to large changes in the input signal of the visual system. The question how we nevertheless manage to perceive invariant object properties has been addressed in many perceptual domains (size, shape,

brightness, color, etc.). Especially in object color perception, constancy has been extensively examined, and widely varying degrees of color constancy have been reported depending, for instance, on the nature of the task and the given instructions (Arend & Reeves, 1986; Radonjić & Brainard, 2016), the choice of experimental stimuli, and also on the individual observer (Hurlbert, 1998; Brainard, 2004; Smithson, 2005; Foster, 2011; Brainard, Cottaris, & Radonjić, 2018). This variability in the degree of constancy was also found in the few studies investigating the case of transparent layer constancy. This holds for the achromatic domain (Gerbino, Stultiens, Troost, & de Weert, 1990; Robilotto & Zaidi, 2004) and also in the general case of chromatic transparent layer constancy (Khang & Zaidi, 2002a, 2002b; Faul & Ekroll, 2012; Faul & Falkenberg, 2015), on which we focus in the present investigation.

In general, we speak of perceptual transparency if we perceive at the same line of sight two different entities, namely a transparent object and the background which is located behind it (Metelli, 1974). This perceptual decomposition of a single local input signal into at least two causes has already been described by Hering (1879, “Spaltung der Empfindung” [Decomposition of the sensation]) as a general concept. In the field of transparency perception, the perceptual decomposition of the stimulus into a transparent layer and a background has been described by the gestaltists (e.g., Moore Heider, 1933; Koffka, 1935) and was later called *scission* (Metelli, 1974; Anderson, 1997). To avoid any unwanted connotation of the term *scission* stemming from its long history and, specifically, to distance ourselves from its meaning as being the inverse of color fusion, we will use the more descriptive term *decomposition* in the following text. Several geometric and photometric/colorimetric conditions influencing the decomposition that accompanies perceptual transpar-

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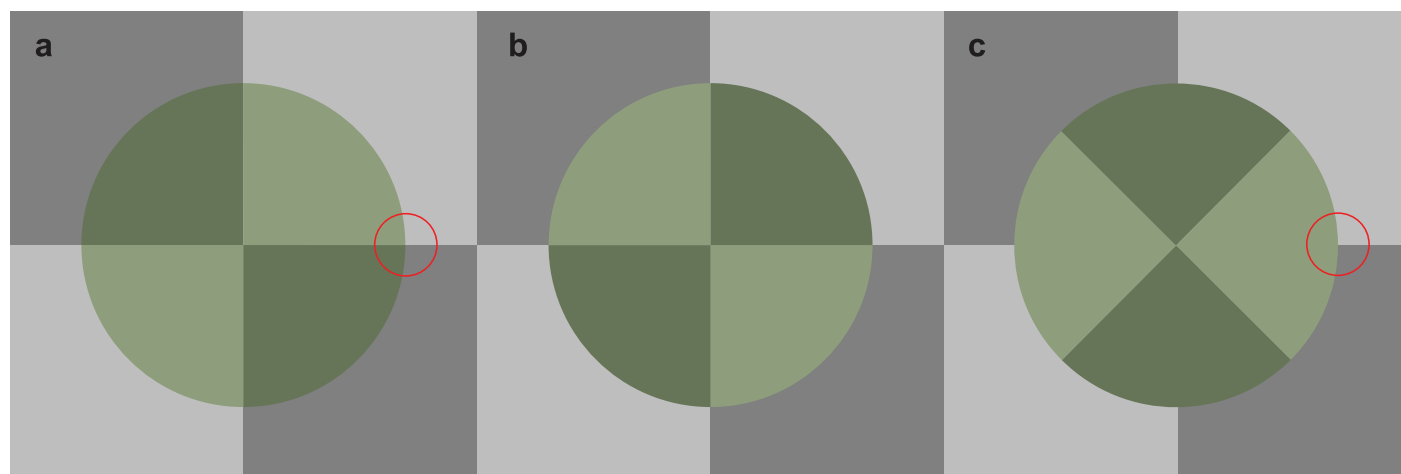


Figure 1. Color relations and figural conditions that determine the transparency impression. (a) Both the colors and the figural properties are selected in such a way that the percept of a green transparent layer is evoked. The red circle marks a so-called *X-junction*. (b) and (c) appear opaque. In (b), the color relations prevent transparency. In (c), the red circle marks a so-called *T-junction*, which is incompatible with transparency. Note that the RGB values of the light and the dark green areas are identical in all four pictures. See text for more details.

ency have been proposed and discussed (Metelli, 1970, 1974; Gerbino et al., 1990; Masin, 1999; Singh & Anderson, 2002; Faul & Ekroll, 2002, 2011; Kingdom, 2008, 2011; Anderson & Khang, 2010).

In Figure 1a, we instantaneously perceive a green transparent layer in front of a 2×2 checkerboard. At the image level, this is partly explained by specific figural properties that support perceptual transparency. Metelli (1974) stressed the importance of the figural unity of the transparent region and the continuity of the boundary lines belonging to the background regions. A typical figural element are X-junctions, which occur whenever a continuous boundary line of a background region crosses the boundary of the transparent object (Figure 1a). It is therefore reasonable that X-junctions are used as a cue for perceptual transparency. This is supported by the finding illustrated in Figure 1c that transparency is usually not perceived without X-junctions (but see Kersten, 1991; Ekroll & Faul, 2013). However, X-junctions alone are not sufficient to elicit an impression of transparency (Figure 1b).

In addition to figural constraints, there are also constraints concerning the four colors at X-junctions. There exist different models that aim to describe the relationships between filtered and unfiltered colors that support a transparency impression (Metelli, 1974; Chen & D’Zmura, 1998; Westland & Ripamonti, 2000; Faul & Ekroll, 2002, 2011). In this work, we refer to the psychophysical filter model suggested by Faul and Ekroll (2002, 2011). This perceptual model is closely related to a physical filter situation (for a brief description of the related physical filter model, see Appendix A). Stimuli generated using the physical

model can be well approximated by the psychophysical model, and the appropriate filter parameters of the latter can be estimated directly from the colors of the physically generated image (for a more detailed description of the model and the parameter estimation, see Appendix B; for an approximately perceptually uniform representation of these parameters, see Faul, 2017). It has been shown that the psychophysical filter model predicts the color conditions for perceptual transparency more accurately than other extant models of perceptual transparency (Faul & Ekroll, 2002). Computer simulations with realistic illumination and reflectance spectra revealed that the filter parameters of the psychophysical model that are estimated for a fixed filter from a physically rendered image remain almost constant when the illumination spectrum and the background reflectances are changed (Faul & Ekroll, 2011).

Different results were reported regarding the question whether color constancy and transparent layer constancy are two different phenomena or whether filter constancy is just a special case of color constancy. Khang and Zaidi (2002a) found similar degrees of constancy for filter objects, regardless of whether valid figural cues were present or not and concluded from this observation that an impression of transparency is not a necessary precondition for transparent layer constancy. In Faul and Falkenberg (2015), on the other hand, we found evidence for a fundamental difference between the two phenomena, since we observed considerably higher degrees of constancy when the figural conditions supported perceptual transparency than with invalid figural conditions that prevented an impression of transparency.

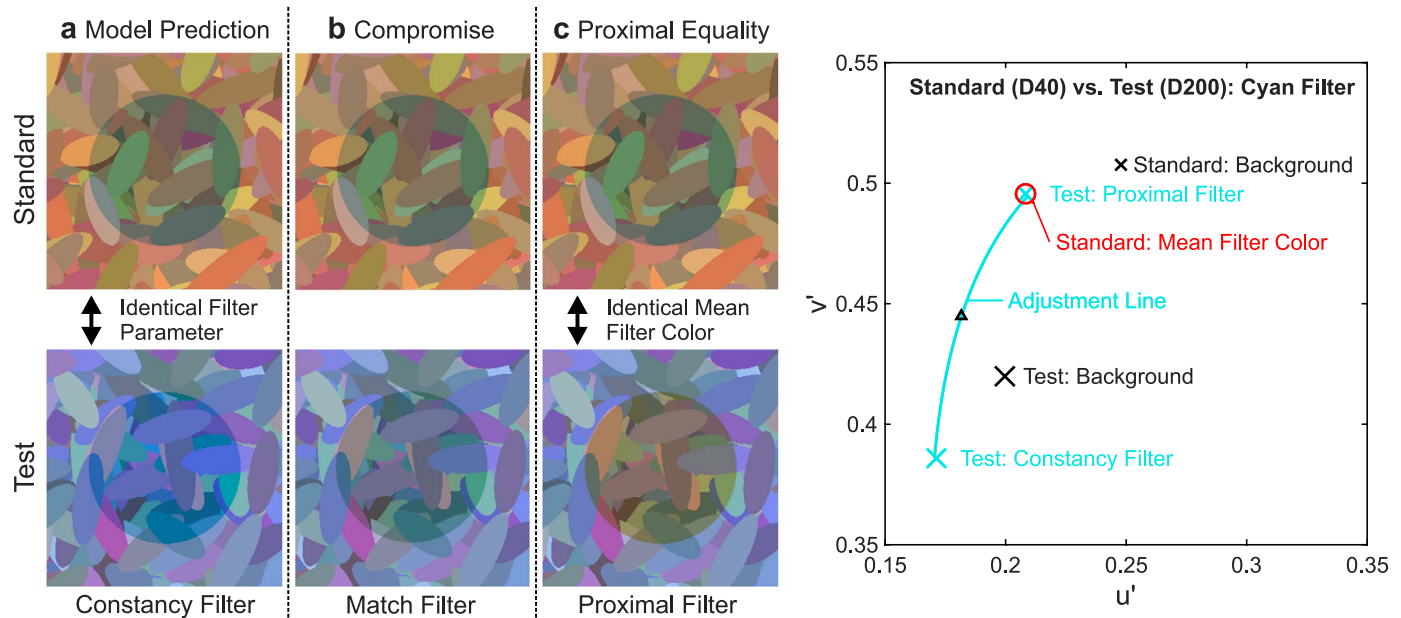


Figure 2. Visualization of the model prediction of constancy (a), the proximal equality criterion (c), and a possible compromise between the proximal and the constancy criterion for the given cyan filter (b). The left panel shows the fixed standard situation in the upper row and the test situation in the lower row. Panel (a): The psychophysical model predicts the constancy filter (mean color in the filter region = X_C) that has the identical filter parameters as the cyan filter in the standard. Panel (c) shows the *proximal filter* that refers to proximal equality, where the mean color in the filter region (X_P) is identical to the mean color in the filter region of the standard (X_S). Panel (b) shows a possible *match filter* that represents a compromise between the criteria *constancy* according to the model and *proximal equality*. In the right panel, the positions of the mean colors in the filter region and the background are shown in the $u'v'$ -chromaticity diagram. The small and large black crosses indicate the mean background color of the standard and the test, and the small and large cyan crosses indicate the mean colors of the same backgrounds seen through the cyan filter. Note that the mean filter color in the standard is identical to the mean filter color in the test for a proximal match. Between the color X_C of the constancy filter and the color X_P of the proximal filter lies the adjustment line, which represents a weighted average of these two colors at each point (see Procedure for calculation details). The open triangle marks the position of the match filter shown in (b).

At first glance, the physical situation underlying transparent layer constancy seems much more complicated than the situation to which color constancy refers, since the ambiguity of the local image signal becomes even greater when, in addition to the illumination and surface reflectance, the transparent layer is added as a third contributing factor. However, whereas in the case of color constancy the ambiguity how a single color should be decomposed into an illumination and a reflectance component cannot be resolved without additional assumptions, this problem does not hold with respect to the estimation of the filter parameters, if one refers to the psychophysical filter model (Faul & Ekroll, 2002, 2011). The reason is that the filter parameters are determined by the color of the background regions seen directly and the color of their covered counterparts, which in the stimulus often lie directly adjacent to each other. For clear filters (i.e., filters without direct reflection; see Appendix B for details) an estimation of the illumination is not even necessary. Thus, according to this model, high degrees of constancy would be expected.

The available findings on the degree of transparent layer constancy are heterogeneous. In particular, different degrees of transparent layer constancy were reported for different types of tasks: While Khang and Zaidi (2002a) found almost complete constancy in an identification task, the constancy performance was significantly limited in cross-context matching (Faul & Ekroll, 2012). Later findings (Faul & Falkenberg, 2015) suggest that the high degree of constancy found in the identification task could be attributed to the specific choice of distractors used in Khang and Zaidi (2002a). Under comparable conditions, a filter that represents a compromise between proximal equality and complete constancy is the preferred choice in both tasks. Figure 2 compares a filter representing transparent layer constancy according to the perceptual filter model with a filter corresponding to proximal equality for the same standard situation. It also gives an example for a possible compromise between these two criteria. Such a compromise has been frequently observed in the past and is also known from other perceptual domains (Thouless, 1931a, 1931b; Faul & Ekroll, 2012; Faul & Falkenberg, 2015; Radonjić, Cottaris, & Brainard,

2015a). In fact, a color constancy index below 1 often represents such a compromise between proximal equality and equality according to some constancy criterion. This means that in order to predict constancy, it does not suffice to describe the criterion for complete constancy, but one also must answer the additional question how the proximal and the constancy criterion for equality are weighted in the perceptual compromise.

For transparent layer constancy, some factors that seem to influence the compromise have already been identified. One factor relates to the filter itself: If the minimum of the transmittance spectrum of the filter and the maximum of the intensity spectrum of the illumination lie in the same wavelength range, this often leads to relatively desaturated filter colors. But subjects do not seem to expect this desaturation: When they see a strongly saturated filter in the standard situation (caused for example by a red filter in a reddish illumination, where the transmittance spectrum and the intensity spectrum of the light have maxima at approximately the same wavelength range), subjects tend to match the filter colors in a rather proximal way even when the test light is bluish and the same filter would in fact appear nearly achromatic (Faul & Falkenberg, 2015). Khang and Zaidi (2002b), on the other hand, found discrepancies from veridical matches if the filter-background combination led to desaturated filter colors in the standard stimulus. In this case, the matched filters (seen in front of an achromatic background) tended to be less saturated than what would have been expected from veridical matches. At least to a certain degree, subjects tend to preserve the filter hue and saturation of the standard stimulus in the match stimulus. Apart from such interactions between filter and illumination, Faul and Ekroll (2012) found higher degrees of constancy when standard and test were not presented simultaneously but were shown in alternation.

Even the highest degrees of constancy observed in previous studies were far away from complete constancy. A possible reason for this could be that the stimuli in previous studies were chosen in an unfavorable way, which led to an incomplete or ambiguous decomposition into a filter and a background layer. That such a decomposition plays an important role in transparent layer constancy is suggested by the observation that a considerably higher degree of constancy results when valid figural conditions lead to a clear transparency impression than when invalid figural conditions prevent perceived transparency (Faul & Falkenberg, 2015). In the first part of this paper, we report tests of single and combined cues that presumably boost this decomposition and thus may improve both the transparency impression and transparent layer constancy. Our results confirm this expectation.

In the second part, we more closely examine the advantage of an alternating presentation of different illuminations. We explicitly test the explanatory power of adaptation in this context and investigate the potential influence of the perceived plausibility of an illumination change.

Experiment 1: Boosting the stimulus decomposition

As already mentioned in the introduction, one explanation for the relatively low degree of transparent layer constancy observed in previous studies could be an incomplete or ambiguous decomposition of the stimulus into two “causal layers.” With the term decomposition we here specifically refer to the perceptual separation of the local color code into two *qualitatively different components*, namely the background and the transparent layer. A phenomenal correlate of this decomposition is perceptual transparency. If this decomposition were always complete, then transparent layer constancy should also be complete (cf. Gerbino, 2015). This expectation is clearly not fulfilled, since only relatively low degrees of TLC have been found so far. For this reason we consider the possibility of an “incomplete decomposition.” Figure 3 provides two examples that the transparency impression, and the properties of the perceived depth layers may vary considerably with the figural conditions, despite identical photometric relations and identical topological conditions according to Kanizsa (1979). The stimuli shown in the upper row of Figure 4 illustrate that the vividness of the transparency impression and thus (according to the just mentioned interpretation) also the degree of the decomposition can differ depending on the background structure. Furthermore, compared with such static stimuli, the transparency impression is clearly more pronounced if the phenomenal separation of a transparent layer from the background is supported by motion or binocular disparity. In the latter case, the transparent layer appears in a different depth plane with a clear distance to the background. In the following we consider several stimulus conditions that presumably support the decomposition and thus should improve the transparency impression and transparent layer constancy. Some of them have already been realized in previous work, but their effectiveness has not been tested.

Potential influencing factors

Type of junction as a basic figural cue for transparency

It has been proposed that X-junctions in the stimulus serve as a cue for perceptual transparency,

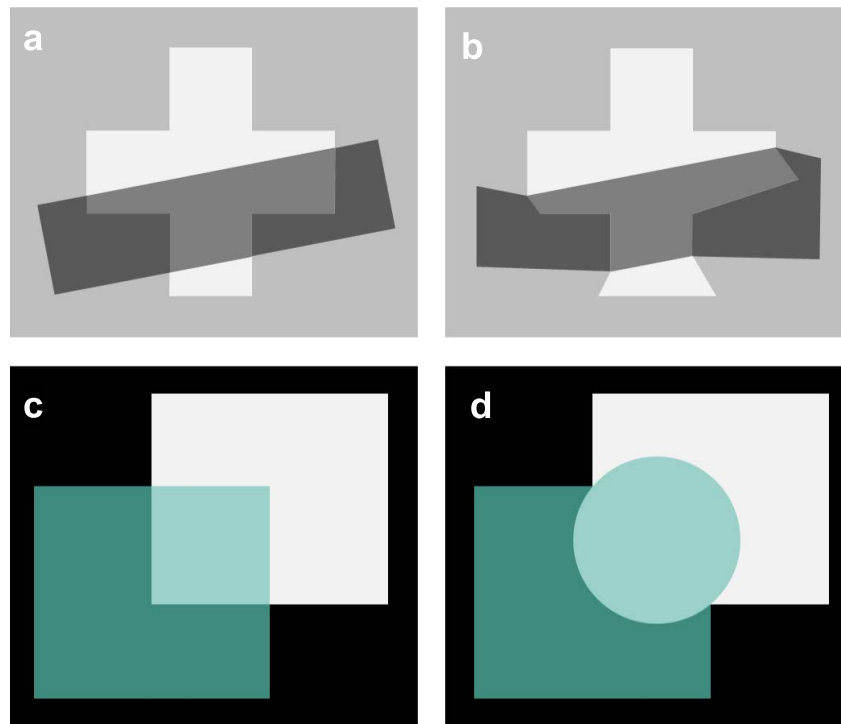


Figure 3. The stimuli in panels (a) and (b), which are slightly modified versions of panels (a) and (c) of figure 1 in Gerbino, 2015, as well as those in panels (c) and (d), are identical with regard to their luminance relations and also fulfill the topological requirements according to Kanizsa (1979), and yet the impression of transparency is clearly weakened in (b) and (d). In panel (a), the impression of a transparent dark bar in front of a white cross is evoked. Panel (b) is ambiguous: The superposing region may also be interpreted as an opaque surface. Note that the perceived colors in the superposed region differ from the ones in (a). Panels (c) and (d) show this effect even stronger for a colored transparent layer; the transparency impression in (d) is clearly reduced or even absent.

since they often coexist with physical transparency. By the term *valid figural condition* for transparency, we refer to the default situation where the transparent layer creates X-junctions with the background. T-junctions, on the other hand, are usually a cue for an occlusion by an opaque object. We will refer to them as *invalid figural condition* for transparency. Of course, these are only heuristics, because X-junctions may also be caused by a nongeneric viewpoint (e.g., Albert, 2001) without physical transparency and, conversely, physically light-transmitting objects can also lead to T-junctions under certain conditions. In addition, how evident it is whether a certain figural condition is fulfilled or not depends on the overall stimulus properties. The basic idea of our experimental manipulation is to reduce the uncertainty of the perceptual interpretation of the investigated objects as *transparent* versus *opaque* by emphasizing the junction type through manipulations of the background and the filter presentation: The better the junction type can be recognized, the stronger the transparency impression should be under valid figural conditions and the clearer the opacity of the object should stand out under invalid figural conditions.

Regularity of the background pattern

It is plausible to assume that in a chaotic background pattern it is much harder to decide whether valid or invalid figural conditions for transparency are present. Therefore, by manipulating the regularity of the background pattern, it seems possible to impede or facilitate the classification of junction types in the scene (Figure 4). To test this prediction, we used two types of background structures in a previous study (Faul & Falkenberg, 2015): The first pattern contained randomly oriented overlapping ellipses, whereas the second pattern was more clearly structured, because it was made of ellipses that were oriented more or less in one direction. In this study, we did not find an effect of background regularity, but since we also used moving filters in all conditions, this might have been due to a ceiling effect caused by filter motion. To investigate this possibility, we here test the effect of the regularity of the background pattern in isolation.

Texture density

Gerardin, Roud, Süssstrunk, and Knoblauch (2006) found that the configural complexity of the background



Figure 4. The upper row shows filter objects with valid figural conditions. The lower row shows the same filter region, but flipped at the horizontal axis, resulting in T-junctions. The regularity of the background pattern influences the recognizability of figural conditions: An irregular background structure (left column) impairs the recognition of the junction type stronger than a more ordered background (right column).

had an impact on the impression of transparency. In their study, configural complexity had two levels, comprising a bipartite field and a 6×6 checkerboard field. The larger number of X-junctions in the checkerboard field led to a more frequent classification of the objects as transparent. One way to increase the number of X-junctions is to increase the texture density of the background pattern (Figure 5). However, this may also have an opposing effect, because shrinking the image structures makes it more difficult to recognize the junction type. This leads to the expectation that texture density and the strength of the transparency impression are related by a reverse U-shaped curve. To verify this assumption, it is necessary to vary texture density in more than two steps. The larger the number of steps, the higher the probability to include the optimal texture density. The prediction for invalid figural conditions is complementary: Perceived opacity should be strongest when multiple T-junctions are unambiguously visible. If the status of the figural condition gets less obvious, the number of misclassifications as a transparent object should increase.

Binocular disparity

Binocular disparity is known as a strong cue that supports a depth segmentation of transparent layers (e.g., Nakayama, Shimojo, & Ramachandran, 1990). Even when the color conditions of a stimulus suggest a certain depth order of the stimulus elements, sufficiently large binocular disparity may override this cue and lead to the perception of an opposing depth order (Trueswell & Hayhoe, 1993). If disparity information indicates that the transparent layer is closer to the viewer than the background and if this depth order is compatible with the depth information indicated by the color conditions, then disparity information should support the decomposition of the stimulus into a transparent layer and the background. This consideration led Faul and Ekroll (2012) to use a stereoscopic presentation of the stimuli when measuring transparent layer constancy in a filter matching task. However, they did not check whether binocular disparity actually had an effect.

Motion

Motion is assumed to facilitate the decomposition by increasing the effectiveness of figural cues for transparency: By continuously covering and uncovering background regions, new X-junctions are created while good continuation of the background pattern and figural unity of the transparent region are preserved in existing X-junctions (Metelli, 1974; Gerardin et al., 2006; Gerbino, 2015). Moving filters were often used with the intention to support the decomposition (D'Zmura, Rinner, & Gegenfurtner, 2000; Khang & Zaidi, 2002b; Khang & Zaidi, 2004; Faul & Falkenberg, 2015) but the effectiveness of this manipulation was not tested. Gerardin et al. (2006) observed almost 100% correct transparency classification in a condition with moving filters. Apart from an effect due to a boosted decomposition, motion could also improve constancy for other reasons: As noted by Khang and Zaidi (2002b), a moving filter generates a larger sample of overlaid background colors than a static filter, which makes it less probable that single surfaces cause a color bias. Moving filter objects also impede a simple proximal match, since it is almost impossible to perform a point-by-point comparison of color appearance.

Alternating presentation

With regard to both color constancy (Foster, Amano, & Nascimento, 2000, 2001) and transparent layer constancy (Faul & Ekroll, 2012) an improved constancy performance has been observed with alternating instead of simultaneous presentation of the two illuminations. Obviously, alternating presentation is not a cue that supports the decomposition of the stimulus into the transparent layer and the back-

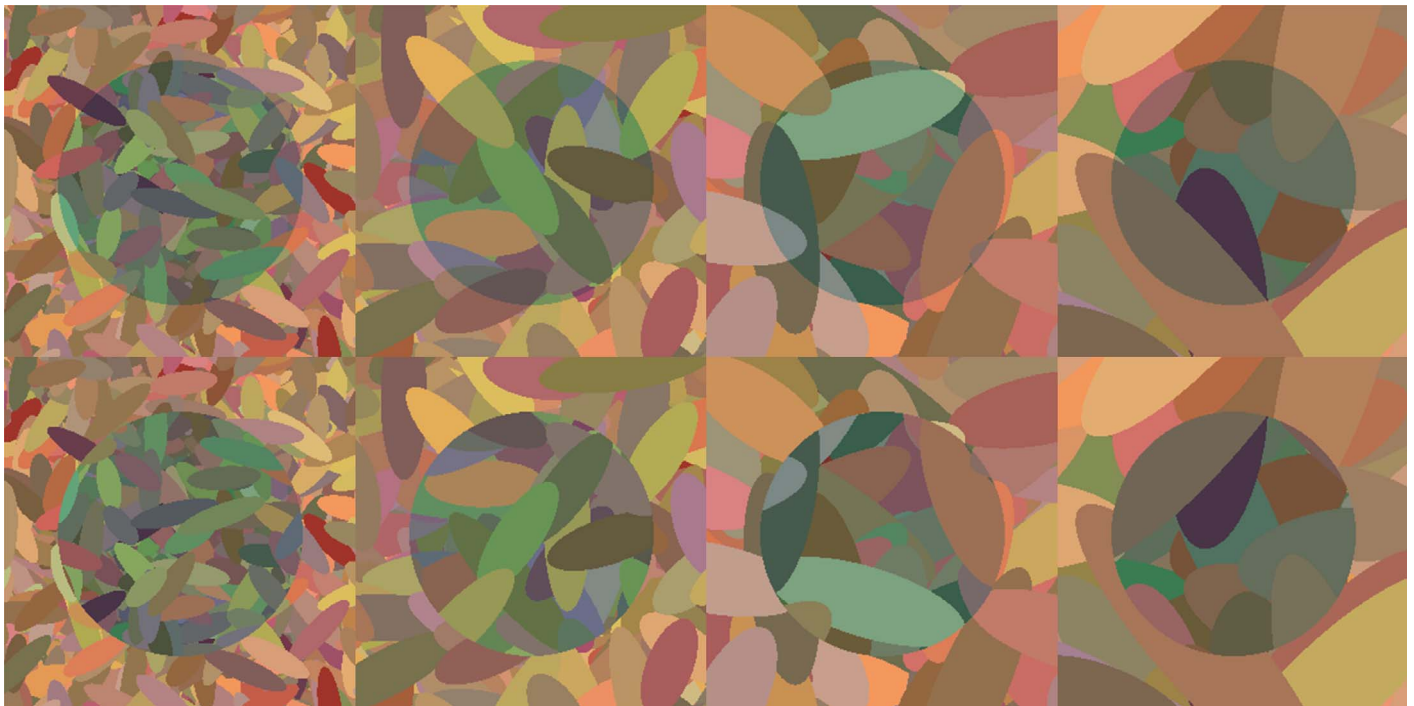


Figure 5. Decreasing texture density of the background pattern from left to right. In the leftmost image, the texture density is so high that the type of a single junction is hardly recognizable. The upper row shows the valid figural condition for transparency; the lower row, flipped filters with invalid figural conditions.

ground. We nevertheless consider it here, because alternating presentation has not yet been investigated with regard to such cues.

Methods

The purpose of this experiment was (a) to quantify the isolated and combined effects of the potential cues outlined above on the transparency impression and the degree of transparent layer constancy and (b) to investigate whether and how these effects are modulated by an alternating presentation. The effects of motion, binocular disparity, background pattern regularity, background texture density, and alternating presentation were determined by comparing each cue condition with a baseline condition, the *minimal cue condition*. In the baseline condition, all tested cues were eliminated: The filter objects were static, had zero binocular disparity, and were shown simultaneously. The background was randomly structured and the texture density was at a medium level (see Figure 6).

We investigated illumination-related transparent layer constancy (Type I constancy, according to Gilchrist et al., 1999), that is, the degree of invariance of the perceived properties of a transparent object against changes in the illumination spectrum, by using an asymmetric filter-matching task across illuminations. In Experiment 1, flat filters were shown in front

of simple Seurat-like backgrounds (cf. Andres, 1997; Mausfeld & Andres, 2002), because such minimalistic stimuli seem especially well suited to investigate to what extent the transparency impression and transparent layer constancy may be increased by the addition of supplemental transparency cues.

General method

Figure 6 shows the stimulus layout of the standard and test stimuli in the baseline condition. The standard filter under the standard illumination was always presented in the upper half of the screen and the test filter under the test illumination in the lower half. In the baseline condition the images for the left and right eye were identical. They had a height of 324.0 mm (35.9°) and a width of 108.0 mm (12.3°).

The background of the baseline stimulus was completely covered by randomly oriented overlapping ellipses with randomly assigned reflectances. All 172 background reflectances were randomly chosen frequency limited spectra (Stiles, Wyszecki, & Ohta, 1977) with a limiting frequency of 1/120 cycles/nm, which were rescaled to a range from 0.0 to 0.8. The resulting smooth and broadband reflectance functions are considered representative for natural reflectance spectra. In the baseline condition, the minor axis of the ellipses was fixed to 10.8 mm (1.24°) and the major axis

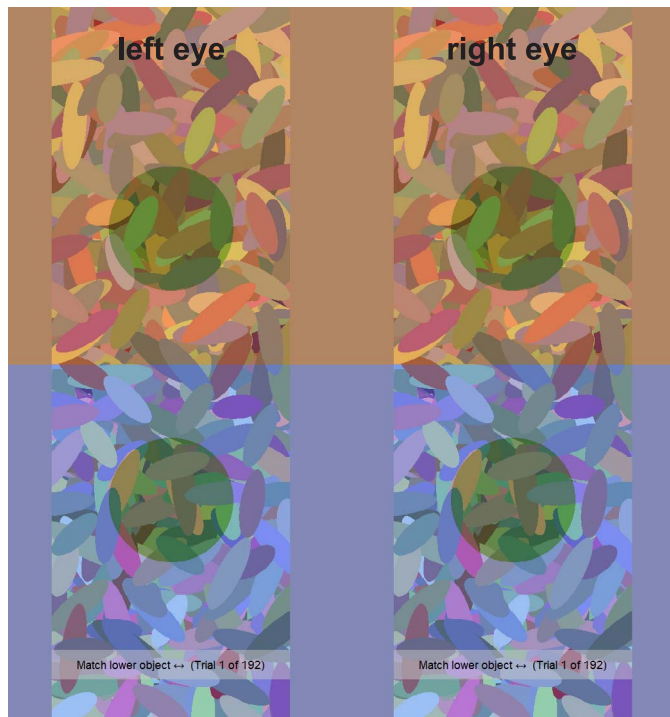


Figure 6. Stimulus layout for the baseline condition of Experiment 1. Both standard filter (top) and test filter (bottom) are shown simultaneously in different illuminations. The color of the uniform surrounding area was identical to the mean background color of the scene, which corresponds approximately to the respective illumination color. The images for the left and right eye were identical, except for the filter in the binocular disparity condition. The test filter shown in this example represents a compromise between the constancy match and the proximal match with a mixing factor of $\alpha = 0.5$ (see Procedure for details).

varied randomly between 21.6 and 43.2 mm (2.48° and 4.95°).

Two CIE daylight illuminations with correlated color temperatures of 4000 K (D40, reddish) and 20,000 K (D200, bluish) were used and both possible illumination changes from standard to test were investigated: D40 to D200 and D200 to D40. To make an illumination change perceptually more plausible, the scene geometry and reflectance spectra of the background ellipses were held constant across the illumination change. In the case of simultaneous presentation, this constancy led to continuous lines in the background geometry at the illumination border. The stimulus was embedded in a uniform surround that had the mean color of the respective background pattern, which corresponds approximately to the respective illumination color.

We used six different transmittance spectra of Kodak CC50 filters (Kodak, 1990) that appear red, blue, green, cyan, magenta, or yellow against a white background. For one of the illumination changes

(D200 to D40) the absorption spectra of some filters were slightly modified to guarantee that all selectable test filters were compatible with the restrictions of the filter model (see Appendix C for details). The diameter of the circular filter objects was 6.49° (56.7 mm).

The light reflected from the background was computed as the pointwise product of the background reflectance and the daylight spectrum. The colors resulting from applying the standard filter were computed according to the physical filter model (Appendix A). These colors do not conform exactly to the psychophysical model, which was found to predict perceived transparency more accurately than the physical model (Faul & Ekroll, 2011). Thus, as a next step, the filter parameters of the psychophysical filter model were estimated from the background colors and the corresponding colors of the filtered background regions in the standard situation (Appendix B). These estimated filter parameters were then applied to both the background colors of standard and test stimulus (Appendix B, Equation 8) to compute the final filtered colors. All spectra used for the calculations were defined from 400 to 700 nm with a step size of $d = 5$ nm. The model parameters were calculated in the LMS color space (Stockman, MacLeod, & Johnson, 1993).

Standard and test stimulus were presented stereoscopically on a calibrated LCD-Monitor one above the other (Eizo ColourEdge CG243W 24.1 in, 1920×1200 pixel, 60 Hz refreshing rate, controlled by a NVIDIA Quadro 600 graphics card with a bit-depth of 8 bit; the monitor spectra used in the calibration of the monitor were measured with the spectroradiometer JETI specbos 1211). In the baseline condition, the two monocular half-images were identical and had a center-to-center distance of 155 mm. The subjects viewed the stereo pairs from a distance of 500 mm through a mirror stereoscope (SA200 Screenscope Pro).

Realization of the cues and experimental design

The default situation is the transparent case with valid figural conditions for transparency (X-junctions). A deviation from the transparent case is characterized by invalid figural conditions (T-junctions). In the invalid figural condition, all the pixels inside the circular filter regions that have been computed for the default situation were flipped upside down at the central horizontal axis. If these invalid figural conditions are recognized, no decomposition into different layers should occur although the color distributions of the flipped objects are unchanged and thus still compatible with a transparency interpretation.

The depth information was varied in two levels: No binocular disparity (*No Stereo*) and binocular disparity (*Stereo*) of the filter regions. The disparity was produced by a nasal shift of the filter region of 2.7 mm

in each half-image (resulting in a 149 mm center-to-center distance of the filter objects). In the *Stereo* condition, the filter objects appeared to float in front of the background. The filters were presented either static or moving. In the *Moving* condition, the filters moved synchronously up and down with a constant speed ($4^\circ/\text{s}$) over a range of 112 mm (12.78°).

The texture density was varied in four levels by scaling the ellipse size e from the baseline condition in four steps: $e/2$, e , $2e$, and $3e$ (Figure 5). Since large ellipses reduce the number of overlaid reflectances, we made sure that there were no repetitions in the ellipse colors in the filter area in those cases. The background regularity was varied in two steps: In the *Random background* condition the orientation of the major axis varied randomly, and in the *Structured background* condition the variation of the major axis was restricted to $\pm 10^\circ$ around a standard orientation of 45° (Figure 4).

In the alternating presentation mode, only the illumination was changed, while the scene geometry was kept constant to enhance the plausibility of the interpretation of an illumination change. Standard and test filter were shown alternately at the same positions as in the simultaneous presentation mode, and each illumination was displayed for 1,200 ms. According to Rinner and Gegenfurtner (2000), more than 80% of the total color adaptation is completed after this period of time.

The combination of these conditions led to a total of 384 stimuli in the first subexperiment: 2 levels of filter motion (*Static* vs. *Moving*) $\times$ 2 levels of binocular disparity (*No Stereo* vs. *Stereo*) $\times$ 2 levels of background regularity (*Random* vs. *Structured*) $\times$ 2 levels of presentation mode (*Simultaneous* vs. *Alternating*) $\times$ 6 filter colors $\times$ 2 illumination changes $\times$ 2 types of figural condition (*Valid* vs. *Invalid*). The second subexperiment, where we investigated the four texture densities for both levels of binocular disparity, contained 192 stimuli: 4 sizes of ellipses (minimal major axis size 10.8, 21.6 (*Baseline*), 32.4, and 43.2 mm) $\times$ 1 level of filter motion (*Static*) $\times$ 2 levels of binocular disparity (*No Stereo* vs. *Stereo*) $\times$ 1 level of background regularity (*Random*) $\times$ 1 level of presentation mode (*Simultaneous*) $\times$ 6 filter colors $\times$ 2 illumination changes $\times$ 2 types of figural condition (*Valid* vs. *Invalid*).

Procedure

We presented the standard filter under one of the illuminations in the upper half of the screen and the test filter under the other illumination in the lower half. The participants' task was to adjust the test filter in such a way that "it looks like the same object seen under the standard illumination." The possible settings of the filter parameters were restricted to a specific adjustment

line that corresponds to a curve segment in color space, which connects a proximal match X_P (i.e., the filter regions in test and standard have the same mean color) and the constancy match X_C (i.e., the filters in test and standard have identical filter parameters). Figure 2 shows an example of the filter colors corresponding to a constancy match, a proximal match, and a possible compromise match, and it also visualizes the adjustment line in color space. The color X_M of a possible match filter is a weighted average of the two colors X_P and X_C and is computed according to the equation $X_M(\alpha) = g^{-1} [\alpha g(X_C) + (1 - \alpha)g(X_P)]$, where $0 \leq \alpha \leq 1$. The function g denotes a transformation of cone excitations to a logarithmized version of the MacLeod-Boynton rb -chromaticity space (MacLeod & Boynton, 1979) and g^{-1} its inverse. This restriction on an adjustment line was introduced, because unrestricted settings of the filter parameters are very tedious and time consuming. This restriction seems unproblematic, because in a previous study with the same task but an unrestricted adjustment of the filter parameters (Faul & Ekroll, 2012), the subjects' settings were always located near this adjustment line in color space. The same restriction has been used in Faul and Falkenberg (2015), and the high ratings of the match quality observed in this study indicate that settings along the adjustment line allow satisfactory filter matches.

The participants could adjust α in the range from 0 to 1 in steps of 0.001 by pressing the left and right arrow keys on the keyboard. Since $\alpha = 1$ corresponds to a constancy match and $\alpha = 0$ to a proximal match, α can be interpreted as a constancy index. In each trial α was initially assigned a random value between 0 and 1. After the best possible setting of α has been made, the participants rated the quality of the match on a five-level scale where "1" denotes a very poor match ("looks like a different object with no similarity to the standard object") and "5" a perfect match ("looks like the same object as presented in the standard stimulus"). In addition, the participants rated the strength of their transparency impression in the adjusted object on a scale from 0 to 10 (where "0" denotes an "unambiguously opaque object, where the background cannot be seen through the object" and "10" an "unambiguously transparent object, where the background is clearly seen through a transparent layer"). The participants also had the option of marking a trial as "invalid" if they noticed that they made a mistake. Each participant completed six practicing trials before the experiment began. Since the task was quite tedious, participants were told to take short rests whenever needed.

Participants

Nine subjects including one of the authors (CF) completed the 384 stimuli of the first subexperiment in

two sessions, each lasting about two hours. In a separate two-hour session, six participants processed the 192 stimuli of the second subexperiment that investigated the influence of texture density. Another three participants including one of the authors (CF) processed 96 stimuli excluding the stereo variation. All participants had normal color vision, as assessed by the Ishihara plates (Ishihara, 1967), and reported normal or corrected-to-normal visual acuity. Participation was voluntary and written informed consent was obtained.

Results

Only 14 out of 4,320 trials were marked as invalid and removed from the data. Due to the large variability in the degree of constancy between the 12 filter-illumination-combinations, which is already known from previous studies (Faul & Falkenberg, 2015), they were treated as independent measurements. In a first step, we checked the effects of the independent variables on the strength of the transparency impression. It was found that objects with valid figural conditions for transparency received relatively high transparency ratings in the majority of cases, even in the baseline condition. The addition of motion, binocular disparity, or a structured background pattern, each led to a significant increase in the transparency rating (see Figure 7, second column). For invalid figural conditions, the addition of these cues had an opposite effect: The transparency ratings were significantly reduced and tended towards zero, i.e., the rate of misclassification as transparent objects was significantly lowered. The size of the ellipses in the background pattern influenced the transparency rating only in the case of invalid figural conditions: The lowest transparency ratings were found with an ellipse size of 32.4 mm (minimum of major axis), indicating that the junction type can best be recognized at this size, whereas for very small ellipses a significant increase in the transparency ratings can be observed. With respect to the transparency ratings, an interaction between all “decomposition-boosting cues” except “texture density” and the figural conditions for transparency is evident.

As mentioned above, the settings of α can be interpreted as a measure of the degree of constancy (see Figure 2 for details on the adjustment line and the Procedure section for details of the calculation of the match filter). Figure 7 (first column) shows the effects of the independent variables on the degree of constancy. The pattern is essentially the same as for the transparency ratings: With valid figural conditions, the addition of motion, binocular disparity, and a structured background pattern leads to significantly higher degrees of constancy compared to the baseline condi-

tion (see Table 1 for statistics). The combined cue condition, in which all three cues were available, led to a cumulative increase in the degree of constancy. Figure 8 compares exemplarily for the cyan and green filter the results of the combined cue condition with those of the baseline condition.

With invalid figural conditions, in contrast, much lower degrees of constancy were observed when the invalid junction type was emphasized by motion, binocular disparity, or a structured background pattern (see Table 2 for statistics). In the combined cue condition, the degree of constancy is strongly reduced, i.e., the test filters were essentially matched in a proximal way.

We conducted a one-way ANOVA for the four realized sizes of background ellipses for both valid and invalid figural conditions. For valid figural conditions there is no effect of ellipse size on the degree of constancy, $F(3, 430) = 0.16$, $p = 0.92$, $f^2 = 0.001$. In the case of invalid figural conditions, the size of the ellipses affects not only the transparency rating but also the degree of constancy, $F(3, 429) = 4.06$, $p = 0.007$, $f^2 = 0.03$. The degree of constancy depending on ellipse size follows a shallow U-shaped curve, with a minimum at 32.4 mm (minimum of the major axis), which corresponds to the position of the minimum of the transparency ratings. These differences, observed at different ellipse sizes, disappear when the stimuli are presented stereoscopically (see Figure A4 in Appendix D).

Figure 9 shows separate scatter plots of the transparency ratings and the respective degrees of constancy for all trials in the baseline and in the combined cue condition. While the transparency ratings in the baseline condition show a considerable overlap across valid and invalid figural conditions, this ambiguity vanishes in the combined cue condition. The correlation between the transparency impression and the degree of constancy was $R^2(429) = 0.37$ ($p < 0.001$) in the full data set comprising both the baseline and combined cue condition.

In contrast to the effects of the decomposition-boosting cues, which differ for valid and invalid figural conditions for transparency, alternating presentation led to significantly higher degrees of constancy for *both* valid *and* invalid figural conditions, and the size of the gain is very similar in both cases (Figure 7, row 5, column 1).

Figure 10 shows the effect sizes for the valid figural condition separately for the single and combined cue conditions and for alternating filter presentation. While the single cues had only relatively small effects on the degree of constancy ($d = 0.25$ to $d = 0.30$), the cumulative effect of the combined cues is substantially larger ($d = 0.70$). The effect of alternating presentation ($d = 0.47$) is smaller than the effect of the combined

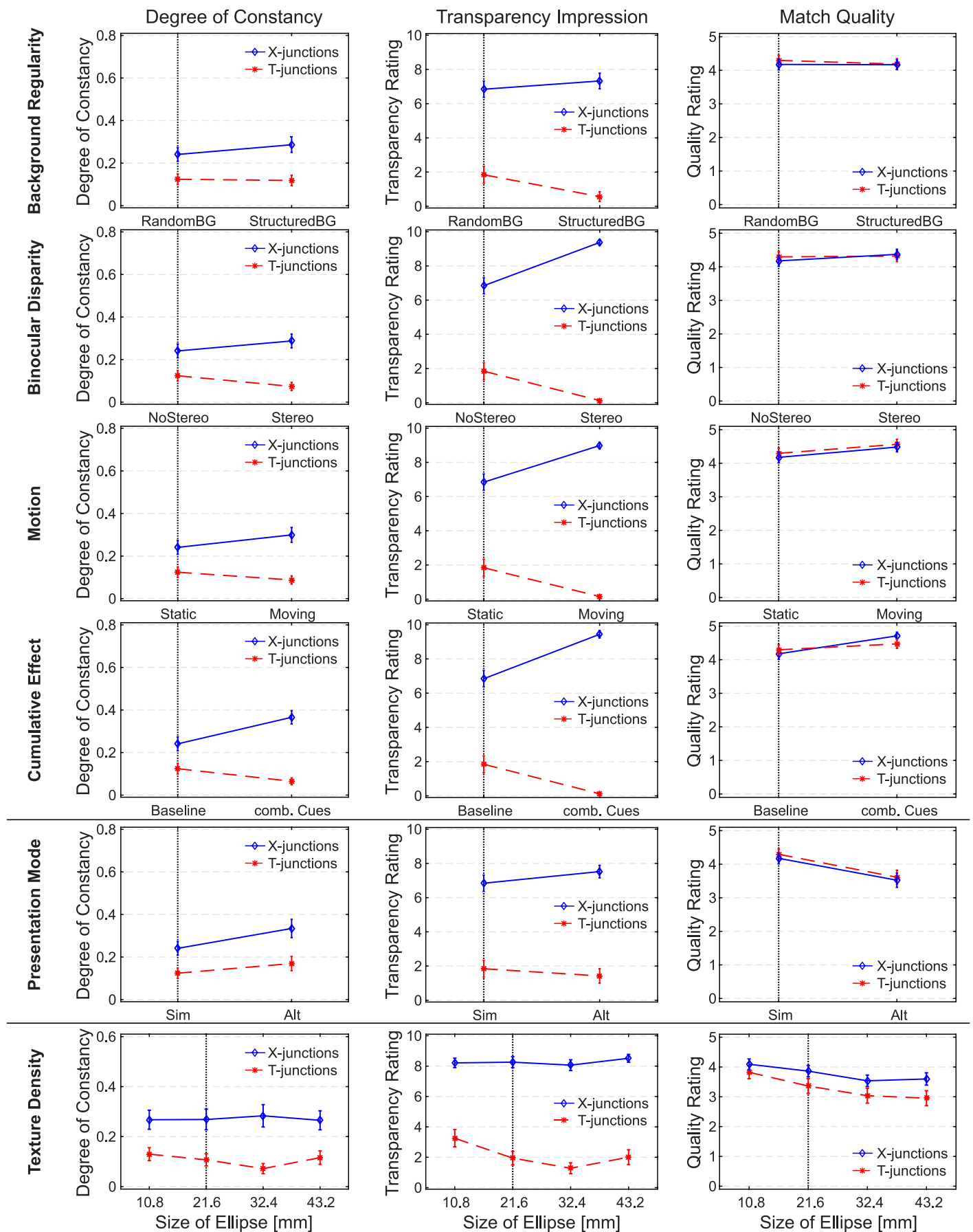


Figure 7. Results of Experiment 1. The rows correspond to the independent variables: background regularity, binocular disparity, motion, the combined cue condition, alternating presentation, and texture density. The columns correspond to the dependent variables: degree of constancy, transparency impression, and match quality, plotted separately for objects with X-junctions (blue open diamond, solid line) and T-junctions (red asterisks, dashed line). Each data point comprises 108 measurements minus any data exclusions. Error bars represent ± 2 SEM. Note that the baseline (marked by a black dashed line) in the texture density plots differs from all the other plots, since other subjects generated the data of this Subexperiment. Also note that the scale for the degree of constancy differs for texture density.

cues, but larger than the effect of each single cue. When alternating presentation is combined with the decomposition-boosting cues, an additional shift to higher degrees of constancy is observed (Figure 10). A gain in the degree of constancy in the combined cue condition can be observed in all 12 investigated filter-illumination-change combinations, but it varies in strength (Figure 11). Note that for the single cues a small tendency in the opposite direction was found in some rare cases.

In the conditions in which two of the three cues were realized to support the decomposition (i.e., stereo + motion, stereo + structured BG, and motion + structured BG), degrees of constancy, transparency ratings, and quality judgments were observed that ranged between those observed in the single cue conditions and in the combined cue condition, in which all three cues were realized (see Tables 1 and 2 and Figures A2 and A3 in Appendix D). Although this also applies to the condition “stereo + motion”, the effect size is considerably smaller than for the two other cue pairs.

The match quality ratings are reasonably high in all conditions (in 80% of the trials it was rated as “4” or “5” corresponding to a “good” or “perfect” match), so it is reasonable to assume that a suitable match was in general found along the adjustment line (Figure 7, third column). The match quality ratings are somewhat lower for the baseline condition, in which the figural

cues are harder to recognize. For both types of junctions, the ratings for static filters were lower than those for moving filters. The match quality ratings were also lower for alternating presentation than for simultaneous presentation. It is remarkable that the perceived match quality decreases with decreasing texture density for both valid and invalid figural conditions in Subexperiment 1b. In general, the match quality ratings for objects with invalid figural conditions are lower than for objects with valid figural conditions.

Discussion

Both the transparency impression and the degree of transparent layer constancy are increased by adding motion, binocular disparity, and a regular background pattern to the baseline condition. This result supports the assumption that these cues contribute to a gradually more complete perceptual decomposition of the stimulus into independent filter and background components. The fact that the combination of these cues led to considerably higher degrees of constancy than each cue alone also supports this conclusion. The effect observed with the combined cues seems not compatible with a *weak fusion* model of cue integration (Clark & Yuille, 1990; Landy, Maloney, Johnston, &

Cue condition	<i>M</i> (<i>SD</i>)	<i>d</i> *	<i>t</i>	<i>df</i> **	<i>p</i>
Baseline	0.24 (0.17)	-	-	-	-
Motion	0.30 (0.18)	0.30	3.17	107	0.001
Stereo	0.29 (0.17)	0.28	2.91	107	0.002
Structured BG	0.29 (0.19)	0.25	2.62	107	0.005
Motion + stereo	0.31 (0.19)	0.32	3.27	107	<0.001
Motion + structBG	0.35 (0.17)	0.62	6.43	107	<0.001
Stereo + structBG	0.31 (0.19)	0.37	3.69	106	<0.001
Combined cues	0.37 (0.16)	0.70	7.19	106	<0.001
Alternation	0.33 (0.22)	0.47	4.75	107	<0.001

Table 1. Degree of constancy (mean and standard deviation) observed for objects with valid figural condition. The effect size *d* as well as the results of one-tailed paired-sample *t* tests refer to a comparison of the respective cue condition with the baseline. Notes: * Cohen's *d* for paired samples. ** The *df* result from pairwise testing of 12 filter-illumination-combinations for each of nine subjects.

Cue condition	<i>M</i> (<i>SD</i>)	<i>d</i> *	<i>t</i>	<i>df</i> **	<i>p</i>
Baseline	0.12 (0.13)	-	-	-	-
Motion	0.09 (0.10)	0.24	-2.51	107	0.007
Stereo	0.07 (0.10)	0.35	-3.66	106	<0.001
Structured BG	0.12 (0.13)	0.03	-0.33	107	0.370
Motion + stereo	0.07 (0.10)	0.37	-3.81	107	<0.001
Motion + structBG	0.09 (0.11)	0.22	-2.37	105	0.010
Stereo + structBG	0.08 (0.10)	0.32	-3.36	107	<0.001
Combined cues	0.07 (0.08)	0.44	-4.59	107	<0.001
Alternation	0.17 (0.18)	0.25	2.58	107	0.006

Table 2. Degree of constancy (mean and standard deviation) observed for objects with invalid figural condition. The effect size *d* as well as results of one-tailed paired-sample *t* tests refer to a comparison of the respective cue condition with the baseline. Notes: * Cohen's *d* for paired samples. ** The *df* result from pairwise testing of 12 filter-illumination-combinations for each of nine subjects.

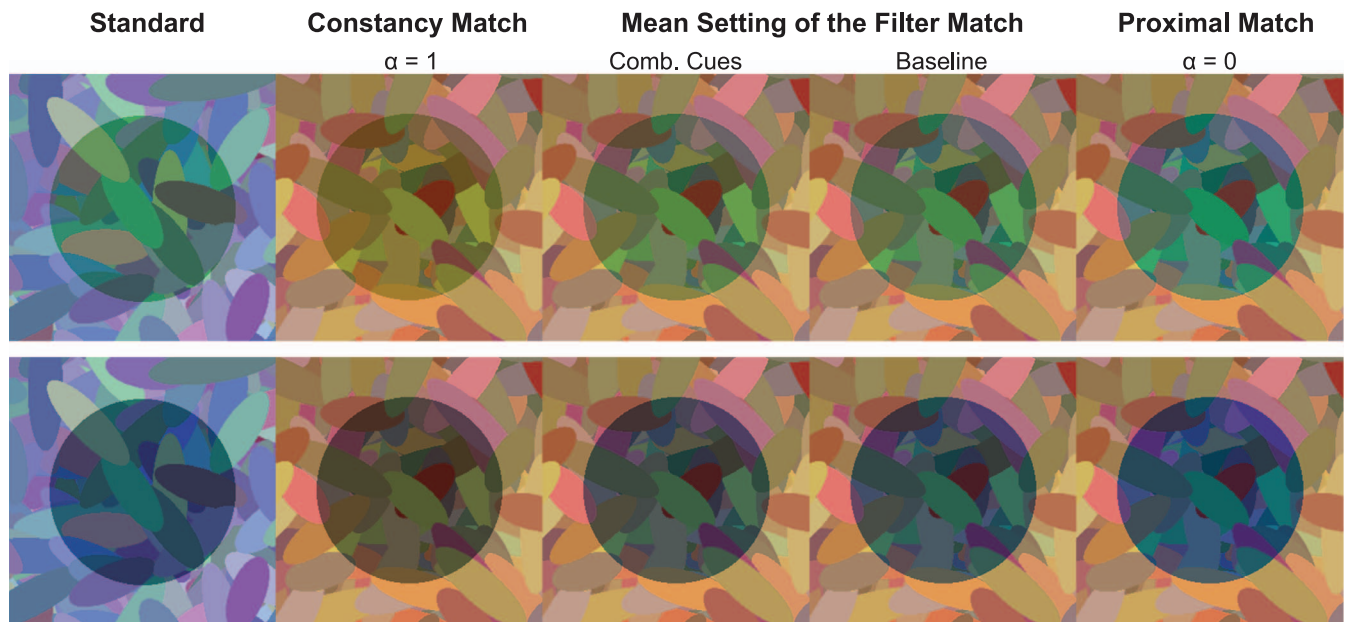


Figure 8. The left images show the green (top) and cyan (bottom) standard filters illuminated by D200. The other images show different match filters illuminated by D40. The second images from the left show the constancy matches (same filter parameters of test and standard filter) according to the psychophysical model. The rightmost images show the proximal match, where the mean color in the filter region is identical in test and standard. The third and fourth images show the mean setting of α in Experiment 1 in the combined cue condition and in the baseline condition, respectively.

Young, 1995; Jacobs, 2002). According to this approach, a weighted average of the individual effects of the available cues would be expected, not the cumulative effect observed here. This indicates that the combined effect of these cues on the layer decomposi-

tion is based on a more complex interaction between them.

While the size of the effect of the isolated cues motion, binocular disparity, and regular background pattern on the degree of constancy was very similar, different effect sizes were observed with respect to the transparency impression: The effect of the regularity of the background pattern was considerably smaller than the effects of both motion and binocular disparity.

Varying the texture density had almost no influence in the case of valid figural conditions, neither on the transparency impression nor on the degree of constancy. With invalid figural conditions, on the other hand, the results correspond to the U-shaped curve, which is expected both for the transparency impression and for the degree of constancy in dependence on the texture density. How can it be explained that noticeable effects were only found with invalid figural conditions, although at a fixed texture density valid and invalid figural conditions should be recognized equally well? In this context, the observation seems important that despite invalid figural conditions, transparency ratings significantly above zero were found, especially at high texture densities. This indicates that if the status of the figural condition is ambiguous, the colorimetric conditions play a superordinate role. This reasoning is in line with the findings of Khang and Zaidi (2002a). However, as soon as compelling violations of the figural conditions for transparency are detected, a veto

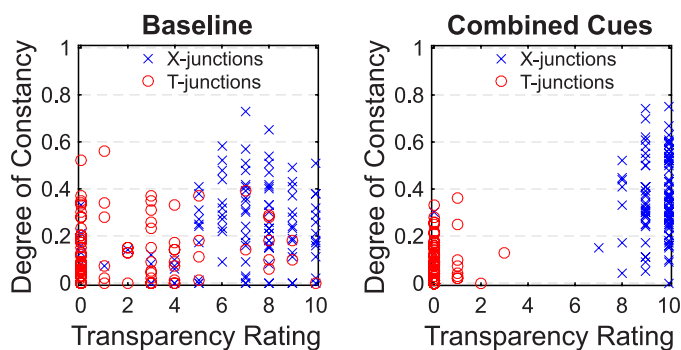


Figure 9. The left panel shows the transparency rating and the corresponding degree of constancy for all data points in the baseline condition for X-junctions (valid condition, blue crosses) and for T-junctions (invalid condition, red open circles). The panel on the right shows all data points observed in the combined cue condition. Whereas in the baseline condition the data for valid and invalid figural conditions show considerable overlap, they are clearly separated in the combined cue condition. The overall correlation between the transparency rating and the degree of constancy is $R^2(429) = 0.37$ ($p < 0.001$).

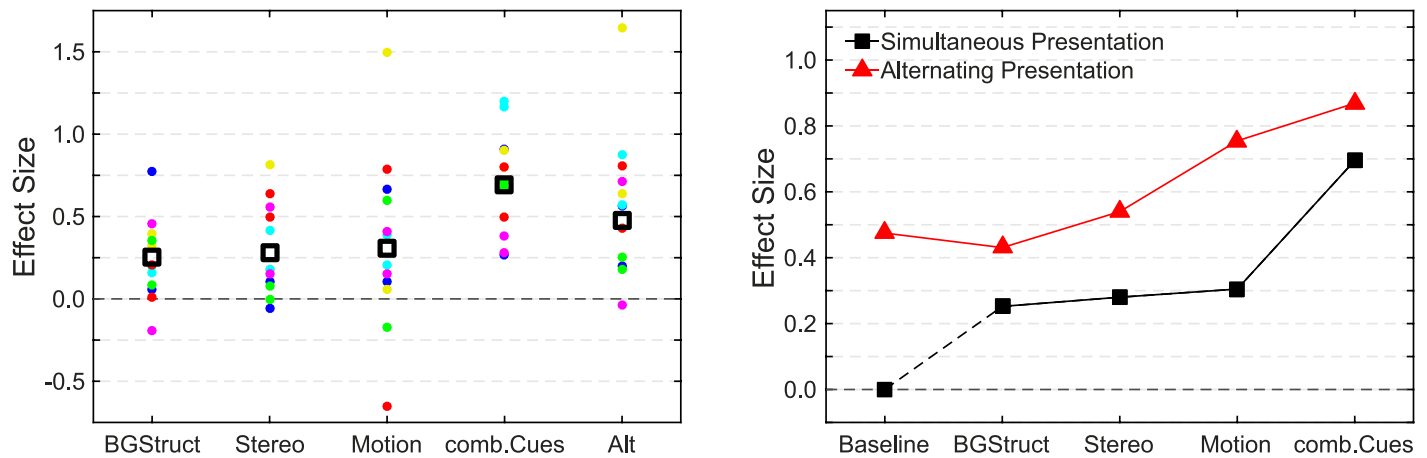


Figure 10. Effect sizes in the valid figural cue condition. In the left panel, open squares indicate the effect size (Cohen's d for paired samples) of the single cues, i.e., background structure, stereo vision, and motion, as well as the cumulated effect of the combined cue condition relative to the baseline condition. For comparison, the effect size of alternating presentation relative to simultaneous presentation is also shown. Each cue condition is compared pairwise to the respective baseline condition ($N = 108$). The dots show the effect sizes separately for all 12 filter-illumination-change combinations, colored according to the respective filter ($N = 9$). The right panel shows the effect sizes for single and combined cues due to enhanced decomposition (black squares) and the combination of enhanced decomposition and alternating presentation (red triangles).

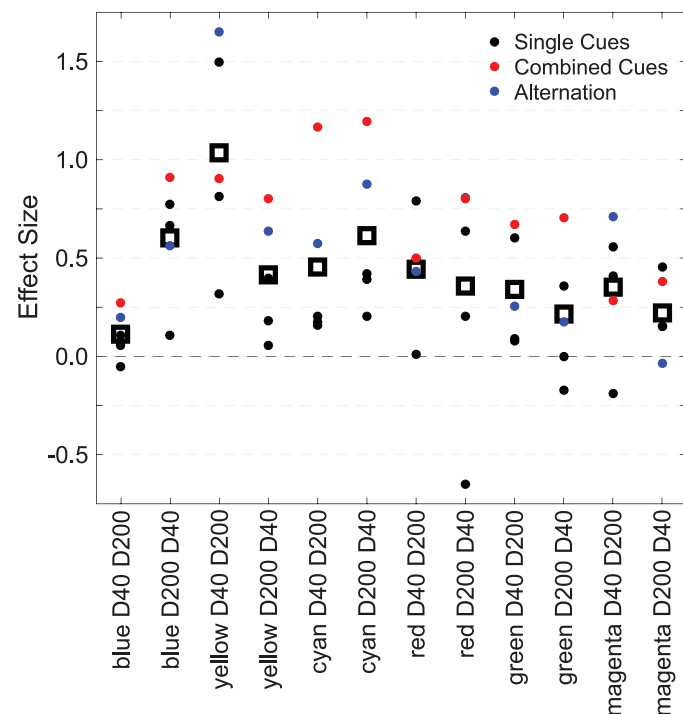


Figure 11. Effect sizes in the valid figural cue conditions depending on the 12 combinations of filter and illumination change. The dots show the effect sizes (Cohen's d for paired samples) for the single cues motion, stereo vision, and background regularity (black dots), for the combined cue condition (red dots), and for alternating presentation (blue dots). The sample size is $N = 9$ in each case. The open squares indicate the average effect size for these five measurements.

against transparency seems to be imposed, and both the transparency impression and the degree of constancy decrease significantly. A variation of the texture density changes not only the number of X-junctions but also the size of the sample of superimposed colors. That the sample size of the stimulus (i.e., its “articulation”) may have an important influence on the degree of constancy is known from the literature on color constancy (Gilchrist & Annan, 2002; Maloney & Schirillo, 2002). Unlike Gilchrist and Annan (2002), we did not find higher degrees of constancy with higher numerosity. This may be due to a ceiling effect because the sample contained already sufficient superimposed colors even with the highest ellipse size used in the experiment.

These results are in line with earlier findings (Faul & Falkenberg, 2015) and show—in contrast to the assumptions of Khang and Zaidi (2002a)—that the degree of layer decomposition has a crucial impact on transparent layer constancy. One could speculate that low detectability of the figural condition might have contributed to the fact that Khang and Zaidi (2002a) did not find an effect of the figural condition on the degree of constancy. However, this is implausible, because we found a clear effect of the figural condition also in the baseline condition, where the figural conditions were identical to those used by Khang and Zaidi. This leads to the conclusion that the main reason for their negative result is to be found in the selection of the distractors in their identification task.

Our results reveal gradual effects of the varied experimental conditions on the degree of perceived transparency and the degree of transparent layer constancy. A natural interpretation of this finding is that

the decomposition of the stimulus into a background and a transparent layer that is thought to mediate these effects can vary in degree. However, the results are also compatible with the assumption that the decomposition occurs in an all-or-none manner and that the variations in the degree of perceptual transparency reflect some kind of compromise between a “decomposition solution” and a “nondecomposition solution” when interpreting the stimulus. It is at present unclear whether and how one could empirically distinguish between these alternative interpretations of the result.

In general, the same cues that presumably boost the decomposition in stimuli with valid figural conditions for transparency have an opposite effect if the figural conditions are invalid, that is, the impression of transparency and the degree of constancy are reduced. In the combined cue condition, where objects with invalid figural conditions for transparency appear unambiguously opaque, the objects were essentially matched in a proximal way. The low degrees of (color) constancy for opaque objects observed in our study are in accordance with the findings of Radonjić et al. (2015a) that the degree of color constancy in simple scenes is much lower (mean CCI 0.10) than in naturalistic scenes (mean CCI 0.47). A possible explanation for this difference is that naturalistic stimuli contain “natural” markers for an illumination change that are missing in simplified scenes. Since such simplified scenes were also used in this study, the low degrees of constancy found for opaque objects with simultaneous presentation might be due to the fact that the change in illumination was not detected. If the illumination was erroneously considered to be uniform, then proximal equality would coincide with equal surface reflectance.

Unlike the cues that facilitate the detectability of the status of the figural conditions and lead to opposite effects for valid and invalid figural conditions, an alternating presentation leads to a similar increase in the degree of constancy in transparent and opaque objects. In the following, we will examine in more detail the mechanisms underlying the gain in constancy due to alternating presentation.

Experiment 2: Alternating versus simultaneous presentation

Experiment 1 confirmed the finding of Faul and Ekroll (2012) that alternating presentation increases the degree of transparent layer constancy. In this second part of the paper, we will consider three explanatory approaches for this enhancement, namely the *changed-strategy hypothesis*, the *plausible-illumination-change hypothesis*, and the *adaptation hypothesis*.

The changed-strategy hypothesis was suggested by Faul and Ekroll (2012), stating that the gain in constancy with alternating instead of simultaneous presentation may be explained by a change in the matching strategy: The visual system usually seems to compromise between the proximal stimulus and a more abstract perceptual representation of the “real” object. Thouless (1931a, 1931b) called this a “phenomenal regression to the real object.” The abstract representation of a real filter object is here assumed to be the *constancy filter*, which is characterized by invariant filter parameters across the two illuminations. With alternating presentation, the standard filter (the proximal criterion) is not present while adjusting the test filter, and the constancy filter might therefore be given a higher weight compared to simultaneous presentation. According to this changed-strategy hypothesis, both criteria would still be present, but their relative influence on the match is modified.

Since the kind of transparent layer constancy considered here refers to a change of the illumination in the environment, we propose another explanation that relates to the perceived plausibility of an illumination change. It rests on the assumption that a constancy prediction of the filter is only relevant to the visual system if a change in the illumination is a plausible interpretation of the stimulus. That a high plausibility of an illumination change might be a core condition for transparent layer constancy is supported by the finding that the perceptual belonging to an illumination “framework” can strongly affect lightness and color appearance (Gilchrist, 1980; Gilchrist et al., 1999; Radonjić, Todorovic, & Gilchrist, 2010). Similar to the changed-strategy hypothesis, this plausible-illumination-change hypothesis postulates an influence on the compromise between the proximal and the constancy criterion of filter similarity, whereby the alternating presentation should be the more effective the more plausible an illumination change appears, given the stimulus conditions.

The adaptation hypothesis, which attributes the effect of alternation on a fast selective adaptation to the two illuminations, appears at first sight rather implausible, because according to the perceptual filter model to which we refer, the estimation of the filter parameters in stimuli without an additive component would only be affected by a *nonlinear type of adaptation*, because otherwise the channel-wise ratios would remain unchanged. However, whether or not adaptations play a role is an empirical question. As just mentioned, even from the perspective of the filter model adaptation might play a role, if a fast nonlinear adaptation process would take place. That adaptation can indeed be fast was shown by Rinner and Gegenfurtner (2000), who found that more than 80% of the total color adaptation following a stimulus change

is completed after the first second. In Experiment 1, an alternation cycle with a duration of 1200 ms for each frame was used. It is thus reasonable to assume that a large part of the adaptation was completed at the end of each presentation period. Simultaneous presentation, on the other hand, provides additional context that could prevent a selective adaptation to the actual illuminations in each region and instead may lead to an adaptation to the overall mean.

Methods

The goal of Experiment 2 was to shed light on the mechanisms that are responsible for the gain in transparent layer constancy with alternating presentation. To this end, three different modes of alternating presentation were investigated, for which the proposed explanatory approaches each predict a different influence on the constancy performance: *FullBin*, *PartBin*, and *PartMon* (see Figure 12 for the corresponding stimulus layouts).

In the *FullBin* mode, the complete scene is alternately shown under two different illuminations while keeping the geometry of the scene constant to plausibly suggest a change in illumination (Figure 12a). The subjects looked binocularly on standard and test stimulus. This mode is identical to the alternation mode used in Experiment 1.

In the *PartBin* mode, also a binocular observation of differently illuminated scenes is used, but in this case only parts of standard and test stimulus are separately shown at distinct locations on the screen (Figure 12b). Compared with the *FullBin* mode, this should considerably reduce the plausibility of an illumination change, since the presented scene is now equally compatible with the alternative interpretation of a systematic change in reflectance.

In the *PartMon* mode, the stimuli and presentation positions are identical to those in *PartBin* mode, but haploscopic viewing conditions are used; that is, each of the two illuminations is restricted to one eye (and in this sense, the presentation is monocular). The advantage of haploscopic matching is that each eye is separately adapted to a distinct illumination (Fairchild, Pirrotta, & Kim, 1994; Bramwell & Hurlbert, 1996; Luo, 2000).

All three explanatory approaches are compatible with an increase in the degree of constancy in each of the three modes of alternating presentation compared to simultaneous presentation. However, they differ in the predictions about specific effects of the three modes of alternating presentation. The changed-strategy hypothesis predicts no difference in the degree of transparent layer constancy between the three investigated modes.

In the *FullBin* mode, the entire scene is geometrically preserved and continuously binocularly visible. Thus, an illumination change is presumably more plausible than in *PartBin* mode, where only parts of the scenes at distinct locations are seen at a time. In the latter mode, it should therefore be more difficult to decide whether the change occurred only in the illumination, only in the background, or in both. The *FullBin* mode should therefore be more beneficial for transparent layer constancy than the *PartBin* mode, provided the assumption of the plausible-illumination-change hypothesis is correct.

According to the plausible-illumination-change hypothesis, no difference is expected between *PartBin* and *PartMon*. The adaptation hypothesis, in contrast, predicts that the *PartMon* mode should lead to higher degrees of constancy as the *PartBin* mode, because a successive monocular presentation of the differently lit stimuli while the other eye is blacked out, allows a more complete adaptation to each illumination compared to binocular alternating observation (Eastman & Brecher, 1972; McCann, McKee, & Taylor, 1976).

Experimental design

The general setup and apparatus were identical to those used in Experiment 1. To focus on the influence of the different modes of alternating presentation on transparent layer constancy, we tried to induce an unambiguously transparent (or opaque) impression by making the figural conditions as salient as possible. We therefore used a structured background pattern and moving filters throughout. Binocular disparity was not realizable due to the monocular presentation of the stimuli in *PartMon*. In the control condition, the two filters were shown simultaneously under different illuminations (*Sim*). For all modes of alternating presentation, each illumination was displayed for 1200 ms. The same six Kodak CC filters and the same two illumination changes as in Experiment 1 were tested. With one repetition for each subject, this setup led to 192 stimuli, including 4 modes of presentation (*Sim*, *FullBin*, *PartBin*, *PartMon*) $\times$ 12 filter-illumination-change combinations $\times$ 2 types of figural condition (*Valid* vs. *Invalid*) $\times$ 2 repetitions.

Procedure and participants

The procedure of the matching task and the two subsequent ratings of the transparency impression and the match quality were also identical to those in Experiment 1. Nine subjects took part in Experiment 2, including one of the authors (CF). They processed all 192 stimuli in a single two-hours session. All participants had normal color vision, as assessed by

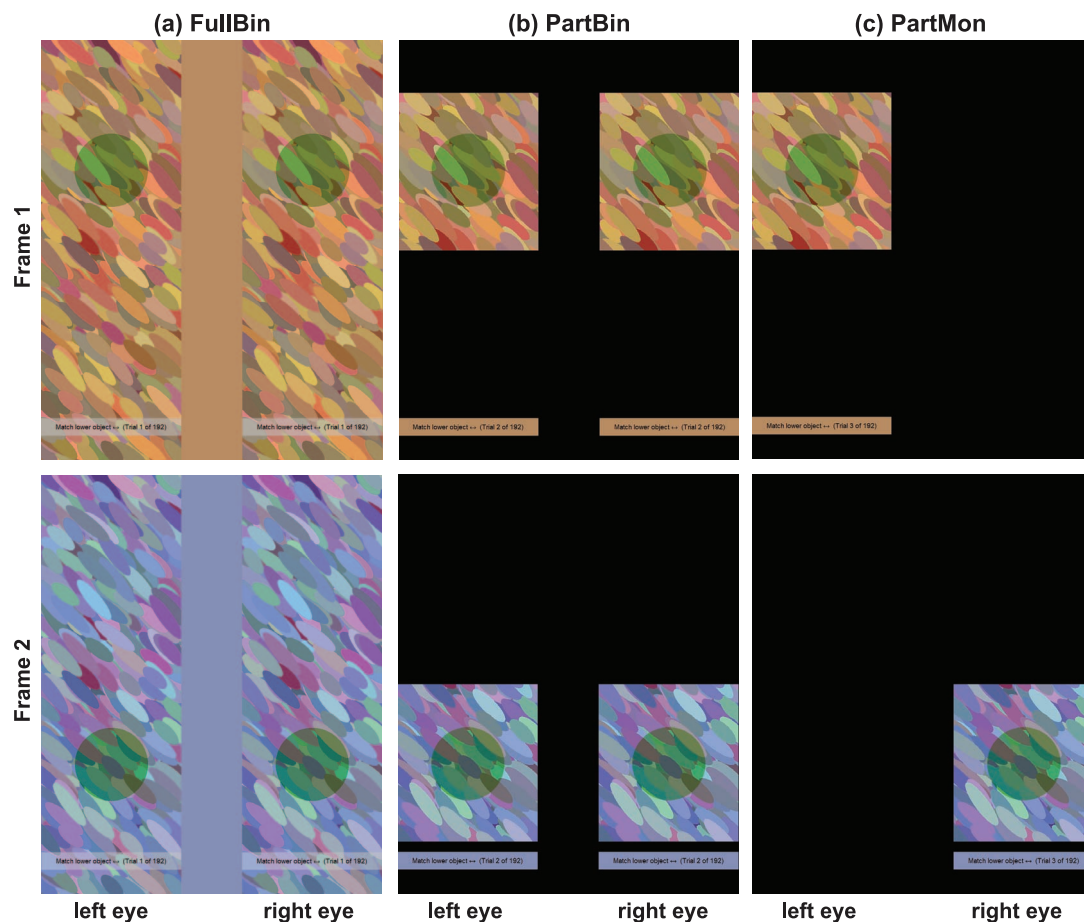


Figure 12. Layout of the stimuli in the three modes of alternating presentation used in Experiment 2. From left to right: (a) Binocular presentation of the full scene with alternating illumination (*FullBin*), (b) binocular presentation of a part of the scene (*PartBin*), and (c) monocular presentation of a part of the scene (*PartMon*). In the upper row the illumination of the first frame is shown and in the lower row the illumination of the second frame (D40 and D200, respectively). The screen is seen through a mirror stereoscope, such that the left side of each screen is projected to the left eye and the right side to the right eye.

the Ishihara plates (Ishihara, 1967) and reported normal or corrected-to-normal visual acuity. Participation was voluntary, and written informed consent was obtained.

Results

Seven out of 1728 measurements that had been marked as erroneous were excluded from the analysis. The mean value of the two measurements per participant was used in the further evaluation ($N = 864$). In one case, no valid measurement was recorded (participant QJ, magenta filter, illumination change D40 to D200, T-junction, *PartMon*). This case was excluded from the other three experimental conditions as well.

In all cases, the transparency ratings are high for objects with valid figural conditions and close to zero for objects with invalid figural conditions. Therefore, it can be assumed that the objects were clearly perceived

as either transparent or opaque (Figure 14). As expected, the transparency ratings do not differ between simultaneous presentation and the three modes of alternating presentation.

Each mode of alternating presentation led to significantly higher degrees of constancy compared to simultaneous presentation of standard and test (Figure 13), both for transparent objects (see Table 3 for detailed statistics) and for opaque objects (see Table 4 for detailed statistics).

For transparent objects the degree of constancy is significantly higher in *PartMon* than in *PartBin*, $t(107) = 3.22$, $p = 0.002$, $d = 0.31$, and also significantly higher than in *FullBin*, $t(107) = 2.49$, $p = 0.014$, $d = 0.24$. In condition *FullBin*, which preserves the geometry of the whole scene, the degree of constancy is slightly higher than in *PartBin*, but this effect is not statistically significant, $t(107) = 1.10$, $p = 0.27$, $d = 0.11$. All results refer to a two-tailed paired-samples t test. Figure 15 shows the effect sizes for all three modes of alternating

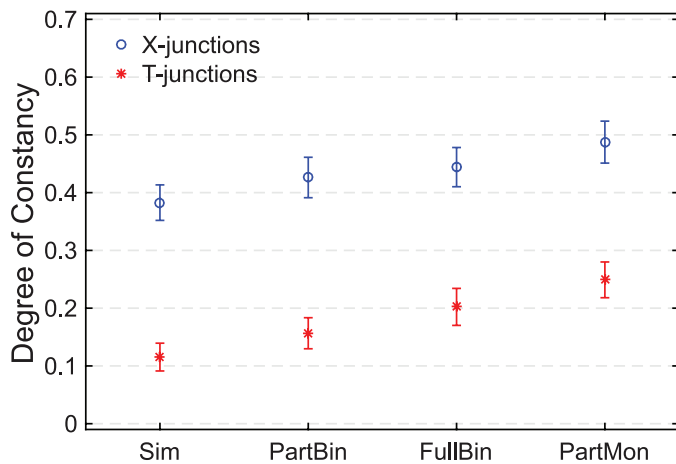


Figure 13. Degrees of constancy observed for simultaneous presentation (*Sim*), and for the alternating presentation modes *PartBin*, *FullBin*, and *PartMon*. Error bars correspond to ± 2 SEM.

presentation for transparent objects and also separate effects for each filter color relative to the baseline condition.

Very similar results were found for the opaque objects: The order of the size of the effect of the experimental conditions on the constancy performance is identical to that found with transparent objects. However, here all three modes of alternating presentation differ significantly from each other: The degree of constancy is higher in *PartMon* compared to *PartBin*, $t(106) = 5.19$, $p < 0.001$, $d = 0.50$, and compared to *FullBin*, $t(106) = 2.63$, $p = 0.01$, $d = 0.25$. The degree of constancy is also significantly higher in *FullBin* compared to *PartBin*, $t(106) = 3.05$, $p = 0.003$, $d = 0.30$. All results refer to a two-tailed paired-samples t test.

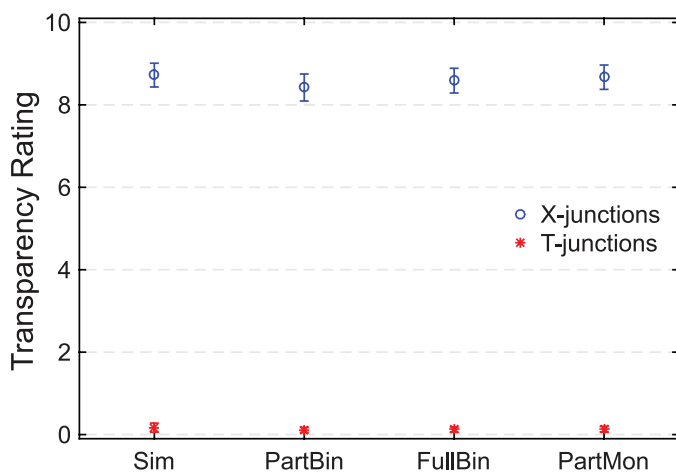


Figure 14. Mean transparency ratings observed for simultaneous presentation, and for the three alternating presentation methods. Error bars correspond to ± 2 SEM.

Presentation mode	<i>M</i> (<i>SD</i>)	<i>d</i> *	<i>t</i>	<i>df</i>	<i>p</i>
Sim	0.38 (0.16)	-	-	-	-
PartBin	0.42 (0.18)	0.30	3.16	107	0.002
FullBin	0.44 (0.18)	0.41	4.25	107	<0.001
PartMon	0.49 (0.19)	0.60	6.20	107	<0.001

Table 3. Degree of constancy (mean and standard deviation) observed for objects with valid figural conditions. The effect size d as well as the results of paired-sample t tests (two-tailed) refer to a comparison of the respective alternation mode with simultaneous presentation. Notes: * Cohen's d for paired samples.

Figure 16 depicts the matching results for the transparent objects obtained with simultaneous and the most effective alternating presentation *PartMon* separately for the two illumination changes realized in the experiment. Shown are the positions of the matching results on the respective adjustment lines in the u^*/v^* -chromaticity-space.

In general, the subjects show different levels of constancy that are consistent across all four experimental conditions (Figure 17). The descending effectiveness of the three alternating modes in the order *PartMon*, *FullBin*, and *PartBin* can be observed quite clearly for six out of nine subjects. For subject AB the value for *PartMon* deviates from this order and for subject QJ the value for *FullBin*. Only for one subject (ND) no effect of alternating presentation was found at all.

The quality rating for the transparent objects was in general reasonable high: Simultaneous presentation led to higher quality judgments (values of 4 and 5 corresponding to a good and a perfect match, respectively, occurred in 84% of the cases) compared to alternating presentation, where a good or perfect rating were given in only 63% of the cases. For opaque objects the quality judgments were lower: 53% good or perfect matches for simultaneous presentation, compared to 41% for alternating presentation (Figure 18).

Presentation mode	<i>M</i> (<i>SD</i>)	<i>d</i> *	<i>t</i>	<i>df</i>	<i>p</i>
Sim	0.12 (0.12)	-	-	-	-
PartBin	0.16 (0.14)	0.26	2.74	106	0.007
FullBin	0.20 (0.17)	0.52	5.40	106	<0.001
PartMon	0.25 (0.16)	0.73	7.57	106	<0.001

Table 4. Degree of constancy (mean and standard deviation) observed for objects with invalid figural conditions. The effect size d as well as the results of paired-sample t tests (two-tailed) refer to a comparison of the respective alternation mode with simultaneous presentation. Notes: * Cohen's d for paired samples.

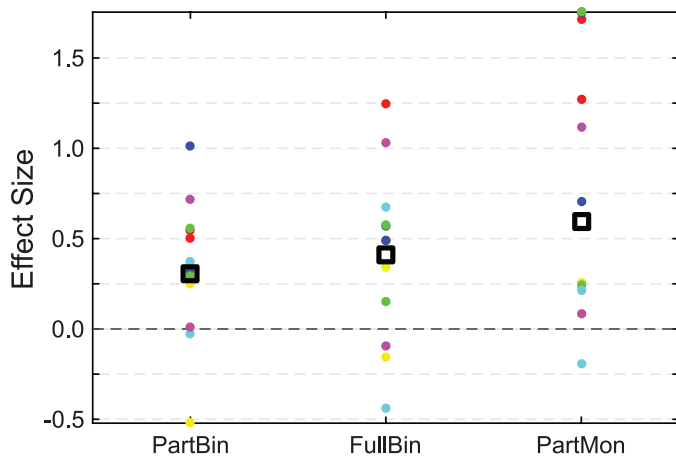


Figure 15. The black open squares indicate the effect size (Cohen's d for paired samples) of the three modes of alternating presentation compared to simultaneous presentation for the valid figural condition ($N = 108$). The dots show the effect sizes for the different alternation methods separately for each of the 12 filter-illumination-change combinations ($N = 9$), the color corresponds to the respective filter color.

Discussion

The observed gains in the degree of constancy with alternating presentation are quite similar to those found in previous investigations. The effect of *FullBin* can be interpreted as a replication of the result of Experiment 1. Faul and Ekroll (2012) reported a comparable increase of the Brunswik ratio between about 0.1 and 0.2 with alternating presentation. In their setting, the scene geometry was not preserved like in *FullBin*, but the stimulus was embedded in the color of the respective illumination. The fact that the effect of alternating presentation in Experiment 2 is somewhat less pronounced compared to the results of Experiment 1 and to those of Faul and Ekroll (2012) is presumably due to the generally higher level of constancy in the present experiment, where decomposition-boosting cues were used throughout. In the domain of color constancy, Foster et al. (2000) found that the color constancy index (CCI) was increased by about 10% by successive presentation of test and standard, which is consistent with our results for the opaque appearing objects. Their experimental setup is comparable to the *FullBin* condition in that they also used constant reflectances and a constant geometry, but in their case

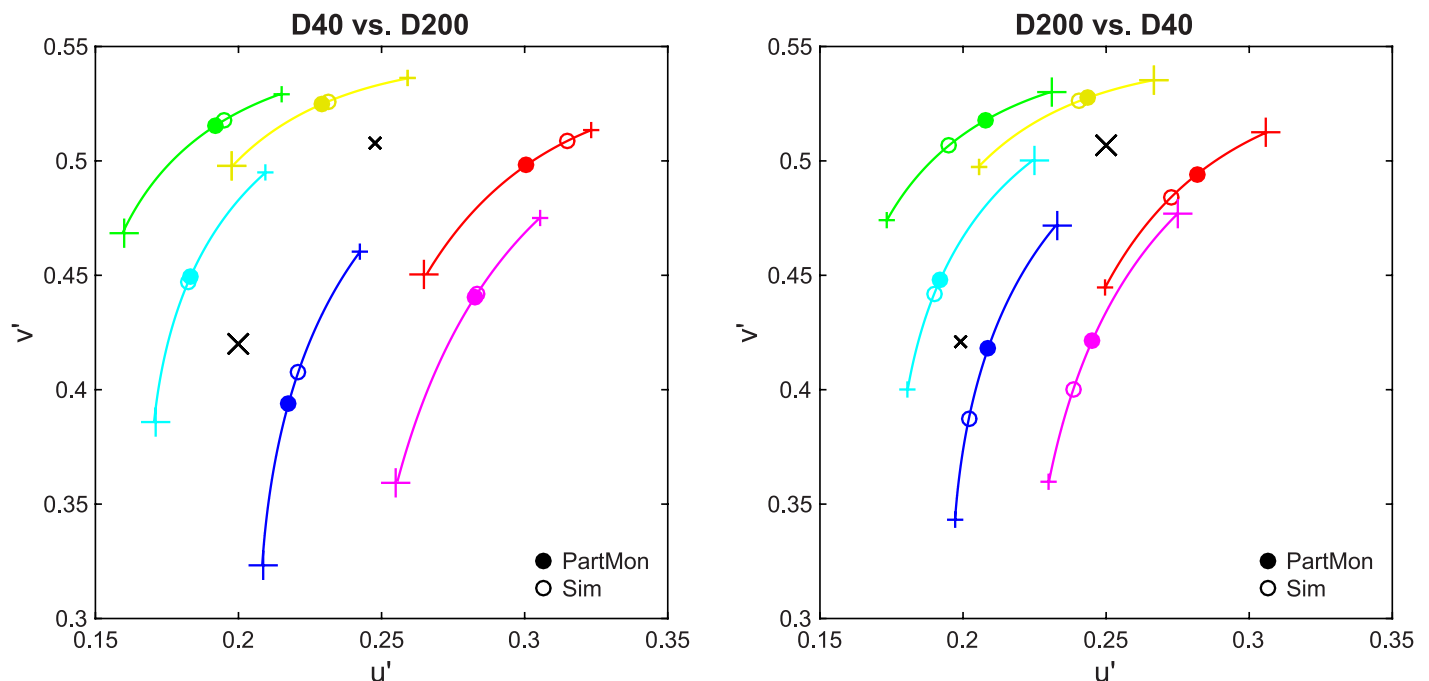


Figure 16. The colored lines show the adjustment lines for each test filter for the respective illumination change. One end of the line corresponds to a *constancy match* (big plus) that refers to a setting of the test filter with identical filter parameters as the standard filter, and the other end corresponds to a *proximal match* (small plus), that refers to the identical mean color in the filter region of test and standard. The mean settings of the filter colors found with alternating presentation mode *PartMon* (filled circles, $N = 9$) are in all cases closer to the constancy match than those found with simultaneous presentation (open circles, $N = 9$), with the exception of the cyan filter in D40 to D200. The black crosses indicate the mean background color in the standard (small cross) and in the test (big cross) stimulus.

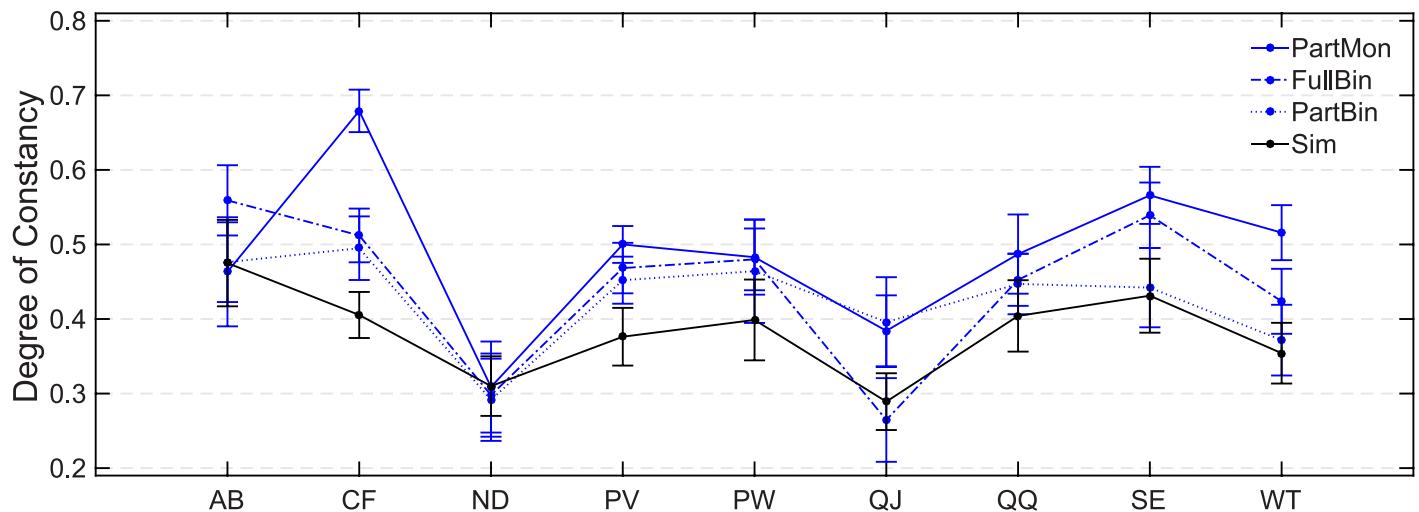


Figure 17. Individual differences in the degree of constancy for transparent objects for all subjects. The pattern of the data for simultaneous presentation (black solid line) is approximately preserved for all modes of alternating presentation (blue lines). Note that each data point corresponds to the mean of 12 measurements for the 12 different filter-illumination-changes combinations. Error bars correspond to $\pm SEM$. The subject identifiers are anonymous codes with the exception of CF, identifying one of the authors.

the target position remained the same, whereas in our setting the positions of test and standard filter were different. These results confirm that both transparent layer constancy and color constancy increase with alternating presentation.

Our results of Experiment 2 suggest that multiple mechanisms contribute to the increase in constancy with alternating instead of simultaneous presentation. The increase of the constancy index with binocular successive presentation in *PartBin* might be explained by the changed-strategy hypothesis proposed in Faul and Ekroll (2012), which states that the strategy used in compromising between the two perceptual criteria changes, if one element—the proximal stimulus—is no longer directly

available for a comparison. But a change in strategy cannot explain the additional gain in constancy that is found with a monocular presentation of the identical stimuli in *PartMon*. The fact that we have found better constancy with haploscopic matching indicates an involvement of a rapid adaptation process. Since the ratios of the filtered colors to the background colors, and thus the filter parameters according to the perceptual model, would not change under a linear gain control, this suggests a nonlinear adaptive color transformation.

Whereas the difference in the results observed in the conditions *PartBin* and *PartMon* can be explained by adaptation, a corresponding interpretation of the difference between *PartBin* and *FullBin* is more difficult. According to the plausible-illumination-change hypothesis *FullBin* should lead to higher constancy than *PartBin*, because a higher plausibility of an illumination change should increase the weight of the constancy criterion relative to the proximal criterion. This hypothesis was confirmed with respect to opaque objects. Although the difference between *FullBin* and *PartBin* was not significant in the filter case, the overall similarity of the effects of an alternating presentation with opaque and transparent objects suggests that there might be a general advantage of *FullBin*. Our results are therefore in line with the assumption that increasing the plausibility of an illumination change benefits constancy.

It is unclear, whether and how the observed superiority of *FullBin* can alternatively also be explained by adaptation: While performing the matching task, the subjects have to look back and forth between the positions of standard and test object on the screen. If

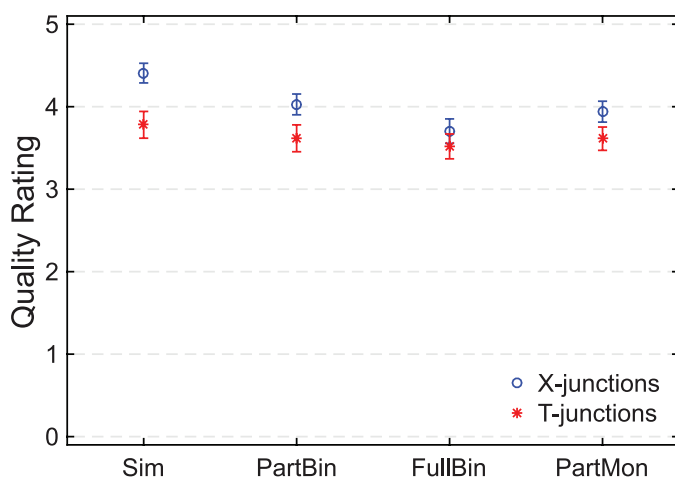


Figure 18. Match quality observed for simultaneous presentation (*Sim*), and for the alternating presentation modes *PartBin*, *FullBin*, and *PartMon*. Error bars correspond to $\pm 2 SEM$.

one only considers the binocularly focused area, which contains the respective object, then there should be no difference between *FullBin* and *PartBin*, because this area is identical in both conditions. But it is very likely that after the stimulus changes, the eyes will stay briefly at their current position before the object is refixated. In this case, the mean color of the new illumination would already be visible in *FullBin* (with unchanged scene geometry), whereas a black background would be visible in *PartBin*. Whether this leads to a faster readaptation in *FullBin* is not clear, because also in *PartBin* a readaptation in the same direction, i.e., toward the neutral point, should start immediately.

To summarize, we found different degrees of constancy in the three modes of alternating presentation. A plausible explanation of this finding is that the settings found with alternating presentation do not only reflect a compromise between a proximal and a constancy criterion, which could be influenced both by a change in strategy as assumed by Faul and Ekroll (2012) or by the plausibility of an illumination change, but that adaptation has also an influence.

General discussion

The aim of this study was to evaluate the contribution of common transparency enhancing cues and presentation modes to transparent layer constancy and to investigate why so far only relatively low degrees of transparent layer constancy were found.

In the first part of the paper, we presented evidence that in stimuli with valid figural conditions the decomposition into a transparent layer and a background can be gradually increased by providing additional cues. Particularly in the combined cue condition, where motion, binocular disparity, and a structured background pattern emphasized the status of the figural conditions, very high transparency ratings were observed for valid figural conditions and very low transparency ratings for invalid figural conditions in otherwise identical situations. As expected, adding these cues to stimuli with valid figural conditions also led to a significant increase in the degree of transparent layer constancy, which supports the assumption that decomposition contributes *gradually* to the constancy performance and that a limited or ambiguous decomposition may have been a problem in previous studies. However, the effects on transparent layer constancy were less pronounced than those on the transparency impression. Further substantial enhancements of transparent layer constancy through a boosted decomposition seem unlikely since the manipulations carried out in Experiment 1 have already led to very strong transparency impressions if the figural condition were valid.

In the second part, we have found evidence that the improved constancy performance with alternating presentation cannot solely be explained by the changed-strategy hypothesis which states that the proximal mode is attenuated because the proximal input of the standard is simply not available during the matching. The further improvement of transparent layer constancy with monocular alternating presentation indicates a supplementary contribution of adaptation. However, the specific nature of the suggested nonlinear adaptation process and the question how adaptation can be taken into account in the filter model remains unclear. Together, these findings suggest that perceptual transparency is not only closely related to simultaneous contrast (Ekroll & Faul, 2013) but also influenced by successive contrast. The results in the opaque case suggest that a higher perceived plausibility of an illumination change has a positive influence on the degree of constancy. A similar trend was found for transparent objects, but in this case the effect was not statistically significant.

The influence of a correct identification of the local illumination framework on the estimation of the constancy prediction

So far, we have assumed that the visual system correctly recognizes the local illumination framework in which the test filter is located, that is, that the perceived illumination framework corresponds to the “real” illumination used in the stimulus generation (Figure 19a). We further assumed that a compromise is formed between a constancy filter calculated in this local illumination framework on the one hand, and the “proximal filter” on the other hand, whereby the weighting of the two criteria depends on the situation. This interpretation leads to the conclusion that the subjects showed only limited transparent layer constancy. However, it is possible that this basic assumption is incorrect and that the actually perceived local illumination framework is more or less unclear.

Traditionally, models of perceptual transparency refer to the four colors at a single X-junction: For this simple case with two background colors A and B, the corresponding filtered colors P and Q can be calculated if the model parameters are known, and conversely, if these four colors are given, the parameters of the respective model can be estimated (Metelli, 1974; Beck, 1978; Westland & Ripamonti, 2000; Faul & Ekroll, 2002). However, Faul and Ekroll (2011) argued that in real scenes this local strategy is ambiguous, because these usually contain several X-junctions with possibly inconsistent color relations and background regions that are completely occluded by the filter object. They

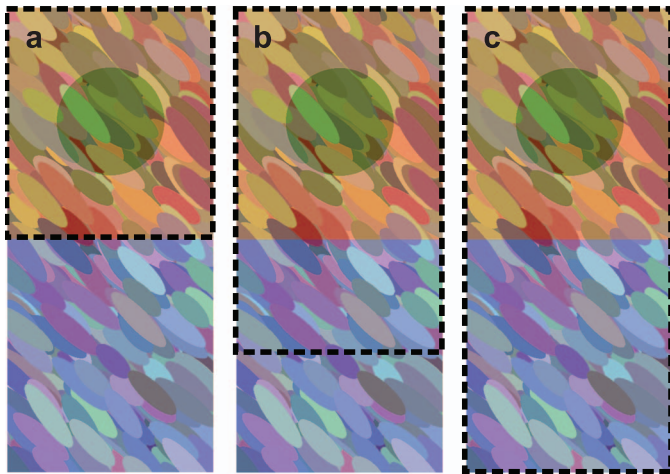


Figure 19. Three hypothetical reference frameworks (surrounded with dashed lines), which imply a different set of background colors for the calculation of the filter parameters. (a) The bi-parted original *local illumination framework* as assumed in the construction of the scene. This situation corresponds to the original calculation of the constancy match. (c) The alternative 100% *global illumination framework*, where a single homogeneous illumination is assumed. In this case the color change in the background is attributed to a change in the background reflectances. (b) Schematic illustration of a partial inclusion of the further context resulting in a weighted mixture of the *local* and the *global* illumination framework.

therefore suggested a more robust estimation of the filter parameters that in the special case of clear filters reduces to the ratio of two colors, namely the mean color of directly visible background regions within a homogeneously illuminated scene and the mean color of the background regions that are occluded by the filter (for details see Appendix B). Since this calculation of the filter parameters obviously depends on the local illumination framework determined by the visual system, it seems worthwhile to consider possible consequences of an erroneous recognition of the illumination framework on transparent layer constancy.

With simultaneous presentation (*Sim*), the different mean background colors of the two halves of the scene are actually due to two different daylight illuminations, but it is unclear whether the visual system interprets the scene in this way and thus whether the background colors of the correct local illumination framework are used in the calculation of the filter parameters (Figure 19a). For example, the integration framework might not end exactly at the intended illumination border, but rather extend beyond it and include other colors from the further context (Figure 19b). In an extreme case, the visual system might assume a single global illumination framework and thus attribute the color changes in the image solely to changes in background reflectances (Figure 19c).

Depending on how observers actually perceive the illumination framework of the filter, different constancy predictions would result according the perceptual filter model (see Appendix E for details). In principle, the local illumination framework could be of arbitrary shape, but for simplicity we here employ a linear blend between the correct local framework and the global framework. The predictions of the constancy filters corresponding to all possible weightings of the local and global frameworks lie on a curve segment in the u^*v^* -chromaticity diagram that is only marginally different from the empirically determined prediction line between the constancy match and the proximal match, that Faul and Ekroll (2012) derived from their data (see Figure 20). This equivalence poses a potential problem in interpreting the observed deviation from constancy, as it might be due either to a limited constancy capability or due to a lack of ability to recognize the intended illumination framework. The latter would result in a different prediction of the constancy filter. If the settings would coincide with this prediction, they would reflect complete constancy, however, with regard to a perceived illumination framework that deviates from the “veridical” one. Figure 21 illustrates the variation of the constancy filter depending on the perceived illumination framework and the two rival interpretations for the observed data in condition *Sim* of Experiment 2.

In order to evaluate the plausibility of these two competing interpretations, we can refer to the conditions with alternating presentation. It is plausible that both spatial and temporal influences of regions outside the local framework reach a minimum in these cases; i.e., the local framework should largely correspond to the veridical one. Nevertheless, we found only limited degrees of constancy. This makes it implausible that the limited degrees of constancy found with simultaneous presentation so far can be explained solely by this *framework-recognition hypothesis*. We can therefore reject the assumption that the observed settings in condition *Sim* reflect complete constancy with respect to an erroneous framework estimation.

The differences found between the three modes of alternating presentation in Experiment 2 clearly show that at least parts of the superiority of alternating presentation have to be traced back to adaptation and possibly also to the perceived plausibility of an illumination change. Like the perceived plausibility of an illumination change, a change of strategy could also affect the compromise between the constancy prediction and the proximal prediction, as proposed by Faul and Ekroll (2012). To what extent the *framework-recognition hypothesis* is suitable to explain the difference between simultaneous and alternating presentation remains an open question. However, the current results already indicate that it cannot com-

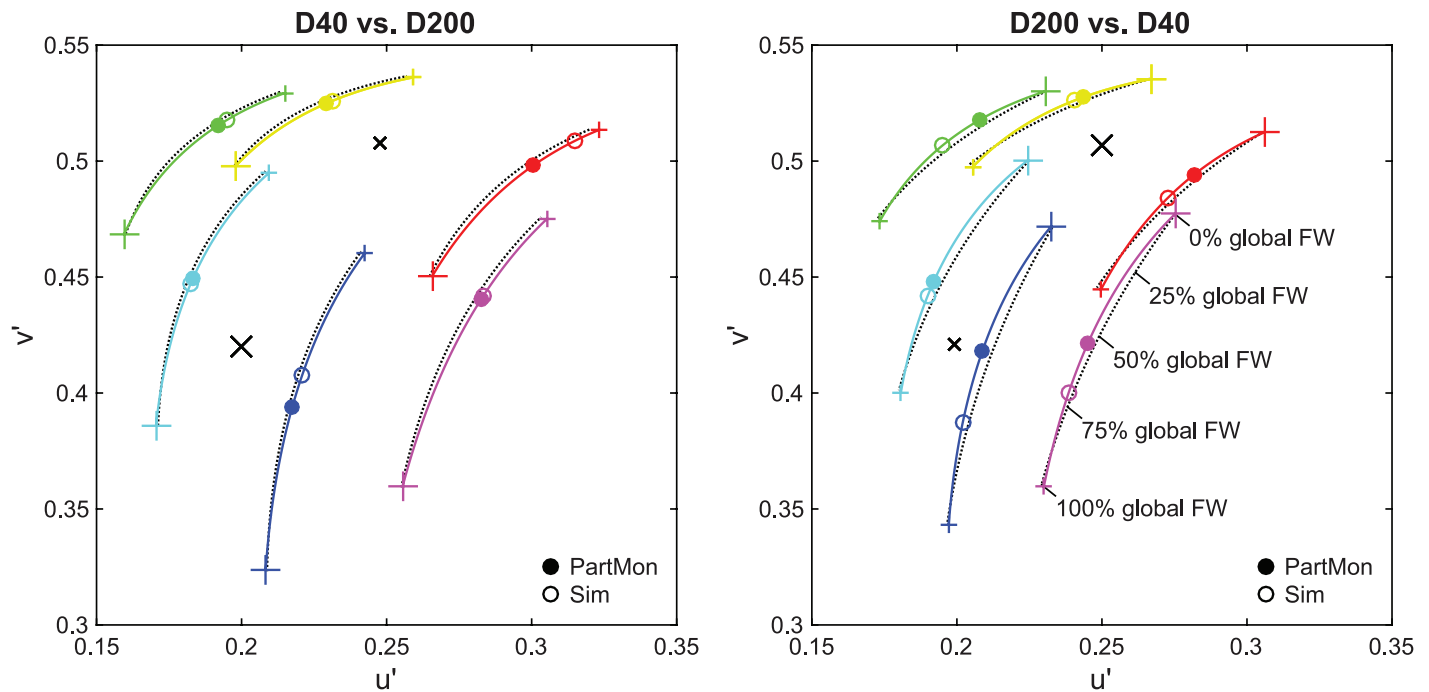


Figure 20. The colored lines show the adjustment lines for each test filter for the respective illumination change. One end of the line corresponds to a *constancy match* (big plus) that refers to a setting of the test filter with identical filter parameters as the standard filter, and the other end corresponds to a *proximal match* (small plus), that refers to the identical mean color in the filter region of test and standard. The black dotted lines show the constancy matches if different framework (FW) compositions for each test filter are assumed. The right panel exemplarily shows the positions of the constancy predictions of the mean filter color for the magenta filter assuming a 0%, 25%, 50%, 75%, and 100% global framework. The “100% global framework” constancy match corresponds to a single global illumination, i.e., all background colors from both halves are included. The 100% global framework constancy match is identical to the *proximal match*. The “0% global framework” constancy match is identical to the *original constancy match* referring to the “correct” local illumination framework.

pletely explain the observed deviations from constancy and probably has only a minor influence. To further elucidate the question in which way the colors used to estimate the filter properties are actually determined, subsequent investigations could systematically examine spatial—and potentially temporal—arrangements of superimposed and surrounding background colors.

Does alternating presentation increase the plausibility of an illumination change?

Different modes of alternating presentation investigated in this work were supposed to influence the plausibility of an illumination change. The question is to what extent this has been accomplished. In one mode of alternating presentation, the geometry of the scenes has been preserved while changing the illumination. Although this condition is clearly conducive to perceive a change in illumination, in abstract scenes like the ones used in the experiment it may still be insufficient to give the vivid impression of a natural change in illumination. It was pointed out much

earlier in other domains of perception that enriched, complex sceneries are beneficial for constancy performance (Holway & Boring, 1941). Radonjić et al. (2015a) investigated color constancy with naturalistic rendered 3D scenes, which naturally featured cues such as illumination edges and cast shadows. Although standard and test object were presented simultaneously in the matching task, the authors have found much higher degrees of constancy than with abstract scenes. The use of naturalistic scenes in a filter situation could be a next step to evoke the perception of two different illumination frameworks within one scene. However, as mentioned by Radonjić et al. (2015a), it would first be necessary to find out exactly what the specific cues are that make our visual system perceive a natural change in illumination. Although there are some suggestions regarding the scene characteristics that typically occur with realistic illumination changes (graduated boundaries, strong luminance variations, orientation of surfaces in space), it is at present largely unclear which of these scene characteristics the visual system actually uses.

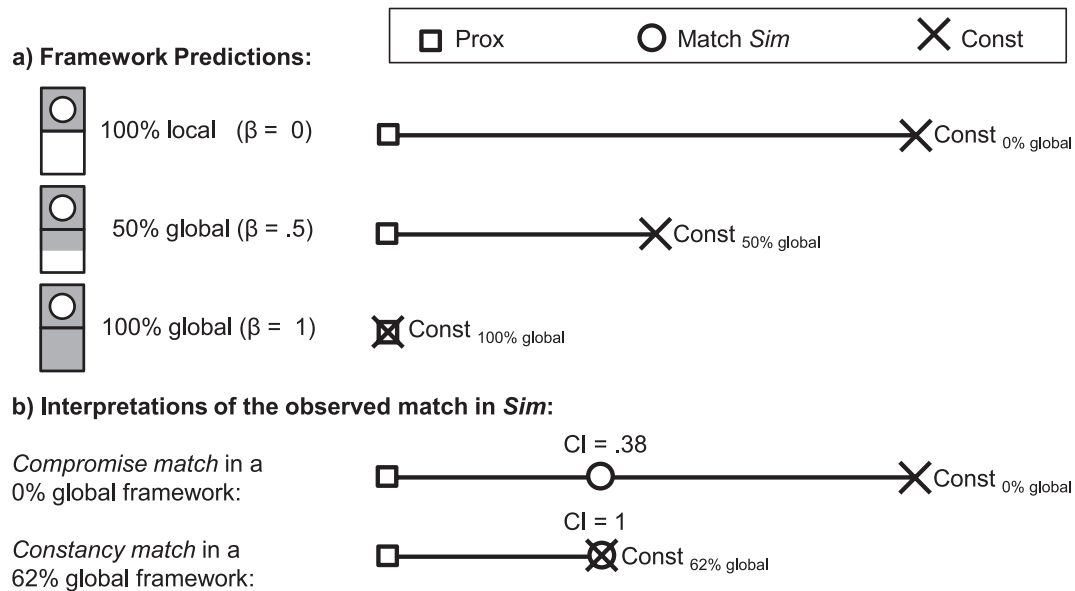


Figure 21. The predictions for the match filter corresponding to complete constancy depend on the relative amount of the background colors of the adjacent stimulus region taken into account (a). If the framework were 100% local—as intended in the stimulus construction—the constancy match would be identical to the original prediction and the locations in color space at which the match settings are expected remain essentially unchanged. They are here presented as a straight line between the proximal match and the constancy match. For a “50% global framework” the constancy match would be different, i.e., closer to the proximal match. For a 100% global framework the predicted colors of the constancy match would be identical to those of a proximal match. (b) There are two extreme interpretations of the observed mean match in condition *Sim*: On one hand, it could be a compromise match with a restricted constancy index (CI) less than 1 with respect to a 0% global framework, i.e., a situation where the local illumination framework is perceived as intended; the alternative interpretation is that adjacent background colors are included to some degree in the estimation of the filter parameters, and that the observed match fulfills the criterion of complete constancy with respect to a 62% global framework ($\beta = 0.62$ is the average of all β values that are closest to the observed matches for all 12 filter-illumination-change combinations).

Relation to color constancy

Contradictory findings were reported with regard to the relation of color constancy and transparent layer constancy. Whereas Khang and Zaidi (2002a) did not find any difference in the degree of constancy for objects with and without valid figural conditions for transparency, we found a clearly larger degree of constancy in the transparency condition (Faul & Falkenberg, 2015). The results of Anderson and Khang (2010) seem also in line with the assumption that perceived transparency may strongly influence the perceived color. They manipulated the relationship of a central patch to its spatial context that induced a transparent or an opaque impression in this target region. In a matching task they found that the perceived color of the central patch deviated more from a proximal match in the transparent than in the opaque case. The present data confirm our previous result that under comparable conditions the degree of constancy is much higher for transparent objects than for opaque objects. This suggests a fundamental difference between color constancy of opaque objects and transparent

layer constancy. But how can this difference be explained?

One reason may be that transparent layer constancy can be more easily achieved than color constancy, if the perceptual properties of the filter are actually determined by the ratio of the background colors to the same colors seen through the filter, as suggested in the perceptual filter model. This is because the information directly available in the proximal stimulus can be used to decompose the color signal into background and filter colors, whereas a similar decomposition of the input signal into an illumination and an object color is not possible without knowing one of the values.

On the other hand, the filter colors predicted by the perceptual filter model in a case without an additive component correspond exactly to the colors that would be expected for an opaque object. This follows from the fact that in image generation the virtual reflectance of a clear filter can also be interpreted as the reflectance of an opaque object (see Equations 2 and 3 in Appendix A). The adjustment line for the investigation of transparent layer constancy can therefore also be interpreted as an adjustment line for color constancy with opaque objects. However, the filter-matching task

used in our experiments does not correspond to a typical experimental set-up for investigations of color constancy, since there was not a single-color patch to be matched, but a whole pattern of different color patches.

If we consider all relations between the adjacent surfaces of the object and the background region in the stimuli, it is apparent that in the transparent case there are considerably more restrictions regarding the color relations between differently colored regions than in the opaque case. This higher degree of indeterminacy in the opaque case also appears to be reflected in the match quality ratings found in Subexperiment 1b and Experiment 2, which were significantly lower than those observed with transparent objects. This result suggests that the participants were more insecure about the actual surface colors of opaque objects than about the more easily accessible filter properties.

Role of the task and instruction comprehension

We observed rather large differences in the degree of transparent layer constancy between individual participants, which is a common finding in color constancy as well (Radonjić, Cottaris, & Brainard, 2015b; Radonjić & Brainard, 2016). It is not yet clear how this interindividual variability can be explained. In research on color constancy, it was found that the instructions strongly influence the matching criteria of the participants (Arend & Reeves, 1986). A possible explanation is therefore that the subjects interpreted the task given in the instruction differently. Depending on the interpretation of the instructions, the perceptual focus may be either on appearance, e.g., on proximal equality, or on more abstract aspects, e.g., on the object's reflectance, which require taking illumination properties into account.

Khang and Zaidi (2002a) came up with the persuasive idea that a more natural task like object identification across illuminations could increase the degree of constancy. However, in a direct comparison of filter matching with filter identification in simple scenes, we did not find any difference in the degree of transparent layer constancy (Faul & Falkenberg, 2015). Radonjić et al. (2015a, 2015b) compared the effect of color matching, simple color selection, and goal-directed object selection on the degree of color constancy in naturalistic scenes and also did not find a difference between these tasks. Radonjić and Brainard (2016) found somewhat lower degrees of color constancy in a matching task compared to a selection task, but only in some of the tested instructional conditions. Since the sample size was not very large, one may argue that this is due to real differences in constancy between the subjects. However, their

additional finding that the subjects changed their settings over the sessions, i.e., either switched to a different perceptual mode or made cognitive changes in task processing, seems incompatible with the assumption of stable traits. To what extent individuals adapt their constancy performance depending on stimulus conditions, task requirements, or explicit instructions remains an empirical question to be answered. However, some results are already available: According to Radonjić and Brainard (2016) instructional effects are significantly lower for naturalistic than for simple stimuli. The available data indicate that certain conditions, in particular simple scenes like Seurat- or Mondrian-like patterns, suggest the use of a proximal equality criterion regardless of instruction and task, whereas in naturalistically rendered scenes, the use of a criterion related to intrinsic object properties like the equality of reflectance seems more natural. Radonjić and Brainard (2016) mention that cognitive processes may influence the behavior of the participants additionally. If one is mainly interested in perceptual processes, it seems therefore reasonable to choose naturalistic scenes and a natural task that exclude such cognitive influences as much as possible.

Conclusions

The present results confirm the conclusion drawn from previous findings that there is a fundamental difference between the perceptual categories of transparent and opaque objects. The investigated cues motion, binocular disparity, and regularity of the background pattern seem to contribute cumulatively to a more complete decomposition of the stimulus into a transparent layer and a background, which suggests a “strong fusion” process in the integration of these cues. Given these results, it is well possible that an incomplete or ambiguous decomposition was one of the reasons for relatively low degrees of transparent layer constancy in previous studies. However, the increase in constancy achievable by improving the decomposition seems rather limited and may have already reached a maximum in the combined cue condition.

In the second part of the paper we investigated more closely the benefit of alternating presentation of standard and test on the degree of transparent layer constancy. The results suggest that the degree of constancy not only increases due to a different compromise between a proximal and a constancy filter resulting from a change of strategy or a higher perceived plausibility of an illumination change, but that adaptation can also have a positive influence.

The fact that the additional effect of alternating presentation is less pronounced if the degree of

transparent layer constancy is already high due to a boosted decomposition indicates that we are already close to the maximum degree of constancy that can be achieved with simple stimuli. It is an open empirical question whether and how the maximum degree of transparent layer constancy can be further increased. A promising path of further research is to consider more natural tasks and naturalistic scenes.

Keywords: transparency, layer decomposition, color scission, transparent layer constancy, color constancy

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Appendix A: The physical filter model

The parameters of the optical filter considered in the model are the absorption spectrum $m(\lambda)$: $0 \leq m(\lambda) \leq 1$, the filter thickness $x > 0$, and the refractive index n . For simplification, a constant refractive index over the wavelengths of visible light is assumed, typically a value between 1 and 2 (e.g., crown glass $n = 1.5$; air $n \approx 1$). If a light ray passes from a medium with a lower refractive index into a medium with a higher refractive index or vice versa, a portion k of the incident light is reflected directly at the boundary surface, while the other portion $(1 - k)$ passes into the second medium. The proportion of direct reflection k is described by Fresnel's equations, with $k = (n - 1)^2 / (n + 1)^2$ (Wyszecki & Stiles, 1982, p.52) for a vertical incidence of light and a medium with a refractive index of $n > 1$. Depending on the absorption spectrum $m(\lambda)$ and the filter thickness x , only a certain portion of the incoming light reaches the opposite side of the filter. This factor $\theta(\lambda)$ is called internal transmission and is given by $\theta(\lambda) = \exp[-m(\lambda)x]$ (Bouguer's law). This process of absorption and reflection occurs recursively and infinitely accompanied by a constant decrease in light intensity. The total reflection $r(\lambda)$ is the limit of the infinite sum of all partial amounts reflected by the filter in the direction of the incident light, i.e., $r(\lambda) = k + (k(1 - k)^2\theta^2(\lambda)) / (1 - k^2\theta^2(\lambda))$. The total transmission $t(\lambda)$ of the filter is calculated analogously from all components passing through the filter, i.e., $t(\lambda) = (1 - k)^2\theta(\lambda) / (1 - k^2\theta^2(\lambda))$.

If a flat optical filter lies coplanar to a background with reflectance spectrum $a(\lambda)$, the light transmitted through the filter is partially absorbed by this background and partially reflected back in the direction of the filter. The remaining light again passes through

the filter, and the amount of light is reduced by reflection components at both filter-air transitions and by the absorption of the filter. The part of the light reflected by the filter in the direction of the observer is the virtual reflectance $p(\lambda)$:

$$p(\lambda) = (t^2(\lambda)a(\lambda))/(1 - r(\lambda)a(\lambda)) + r(\lambda). \quad (1)$$

If the refractive index is $n = 1$, then the direct reflection is $k = 0$, and this implies $r(\lambda) = 0$. In this case the total transmission $t(\lambda)$ is equal to the inner transmission $\theta(\lambda)$, and the formula of the virtual reflectance $p(\lambda)$ simplifies to

$$p(\lambda) = \theta^2(\lambda)a(\lambda). \quad (2)$$

The cone excitation P_i corresponding to the *virtual* reflectance spectrum $p(\lambda)$ under illumination spectrum $I(\lambda)$ is given by

$$P_i = \int p(\lambda)I(\lambda)R_i(\lambda)d\lambda, i = L, M, S. \quad (3)$$

where $R_i(\lambda)$ is the receptor sensitivity for cone type i .

Appendix B: The psychophysical filter model and color calculation of the test filter

We refer to a psychophysical filter model formulated in terms of color codes that was proposed in Faul and Ekroll (2002, 2011). The predictions of this model are given by

$$P_i = \tau_i(A_i + \delta I_i), i = L, M, S \quad (4)$$

where A denotes the color code of a background region, and P the code of the same region viewed through the filter. In the following we will always assume that the color codes are cone excitation values, where the index indicates one of L, M, S . The parameters τ , δ , and I are related, respectively, to the squared filter transmittance $t^2(\lambda)$, the direct reflection factor k , and the illumination spectrum of the physical model. This interpretation motivates the parameter restrictions $0 \leq \tau_i \leq 1$ and $\delta \geq 0$. The filter model reflects both the subtractive character of the physical model of light transmitting objects (as the parameter τ is directly related to the filter transmittance) and its additive character (as the parameter δ represents the direct reflection depending on the refractive index). The parameters can be estimated from $m \geq 2$ background colors A_j and the corresponding filtered colors P_j that result in regions where the filter (partially) overlaps the background. In Faul and Ekroll (2011) robust estimation techniques were proposed that use the mean and

the (uncorrected) standard deviation (std) of these colors. In a first step, τ_i is directly estimated by

$$\tau_i = \frac{\text{std}(P_i)}{\text{std}(A_i)}, i = L, M, S. \quad (5)$$

In a second step δ is integratively estimated via all three color channels by

$$\delta = \sum_i u_i v_i / \sum_i v_i^2, i = L, M, S, \quad (6)$$

where $u_i = \text{mean}(P_i) - \tau_i \text{mean}(A_i)$ and $v_i = \tau_i I_i$.

In the last step τ is finally estimated by

$$\tau_i = \frac{\text{mean}(P_i)}{\text{mean}(A_i) + \delta \times I_i}, i = L, M, S. \quad (7)$$

where the index i refers to different color channels. As a simple estimate of the illumination parameter I , the mean of the background colors A_j can be used. For a model with refractive index 1—as assumed in this paper—the parameter δ is zero and the model reduces to

$$P_i = \tau_i A_i, i = L, M, S, \quad (8)$$

and the unknown parameter τ can be estimated robustly for $m \geq 2$ background colors and the corresponding filter colors by

$$\tau_i = \text{mean}(P_i) / \text{mean}(A_i), i = L, M, S. \quad (9)$$

Note that in this case no estimation of the illumination I is necessary to calculate τ .

In this way, $\tau_{\text{Standard},i}$ can be calculated from the physically generated standard situation. To calculate the filter colors for the test filter, the filter colors Q_i of a constant match are calculated in a next step for the test situation with background colors B_i via $Q_i = \tau_{\text{Standard},i} B_i$. The mean colors in the region of the test filter of a proximal match are equal to the mean color in the region of the standard filter. Thus, the filter parameters τ_{Prox} of a proximal match can be calculated by

$$\tau_{\text{Prox},i} = \text{mean}(P_i) / \text{mean}(B_i), i = L, M, S, \quad (10)$$

Appendix C: Modified absorption spectra

The filter parameters τ_i of the psychophysical model are restricted to $0 \leq \tau_i \leq 1 \forall i$, since τ is related to the squared filter transmittance $t^2(\lambda)$. If very saturated filters are used, values of $\tau > 1$ may occur, when the filter parameters for a proximal match are calculated according to Equation 10 (Appendix B). According to the model, such a filter is not perceived as transparent

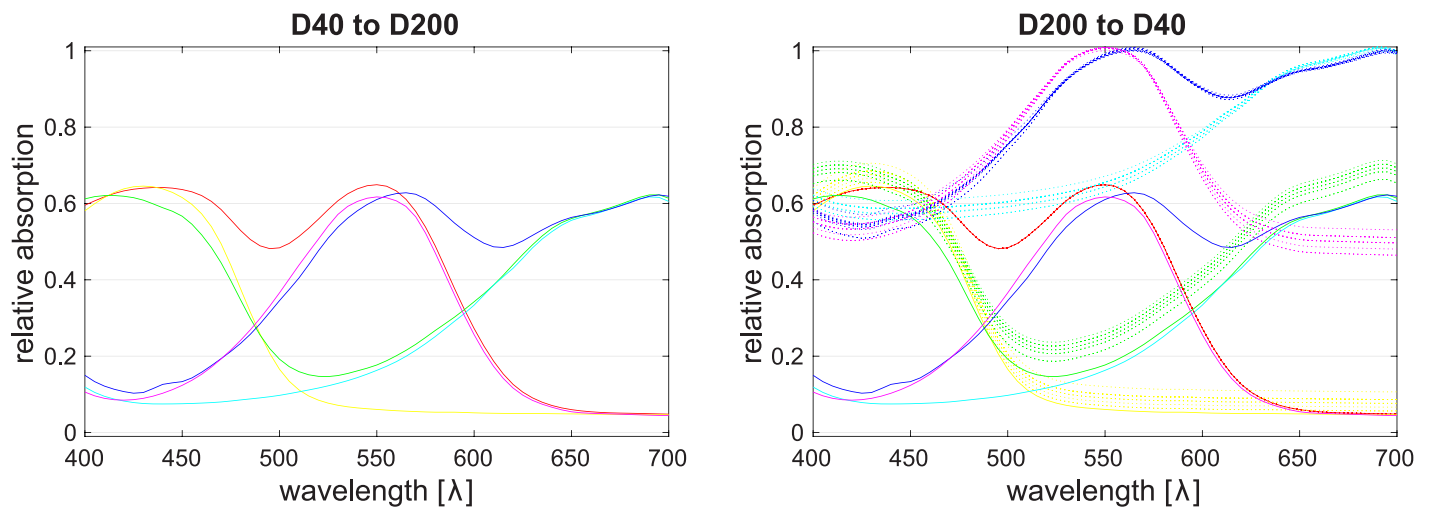


Figure A1. The left panel shows the original CC50 Kodak filters, red, green, blue, yellow, cyan, and magenta used for all background configurations under the illumination change from D40 to D200. The right panel shows the original and the shifted absorption spectra used for the different background configuration under the illumination change from D200 to D40. See text for details.

(in fact, such highly saturated filters often appear luminescent). To prevent this, the relative absorption of the original filters was increased in some cases by the following procedure: If the original absorption spectrum led to τ values greater than 1 under an illumination change, the first step was to shift the absorption spectrum by an additive constant in order to maintain the form of the spectrum. This led to somewhat darker filters. If this was not sufficient to meet the parameter restrictions, the absorption spectrum was additionally flattened by a scaling factor, which led to less saturated filters. This procedure was done individually for all eight background configurations, i.e., the baseline condition and the three types of

pattern density of the background, in each of the two degrees of background regularity. Under the illumination change D40 to D200 no adjustment was necessary. Under the illumination change D200 to D40 the absorption spectra of the yellow and the green filter were shifted only slightly (plus 0.01/0.08 and 0.04/0.12 respectively) while for the cyan, blue, and magenta filter the shifts were considerably larger (plus 0.45/0.5, 0.39/0.42, and 0.41/0.47, respectively). Scaling was necessary under the illumination change D200 to D40 for the cyan, blue, and magenta filter with scaling factor (minimum/maximum) 0.69/0.83, 0.76/0.95, and 0.77/0.95, respectively.

Appendix D: Supplementary data Part 1

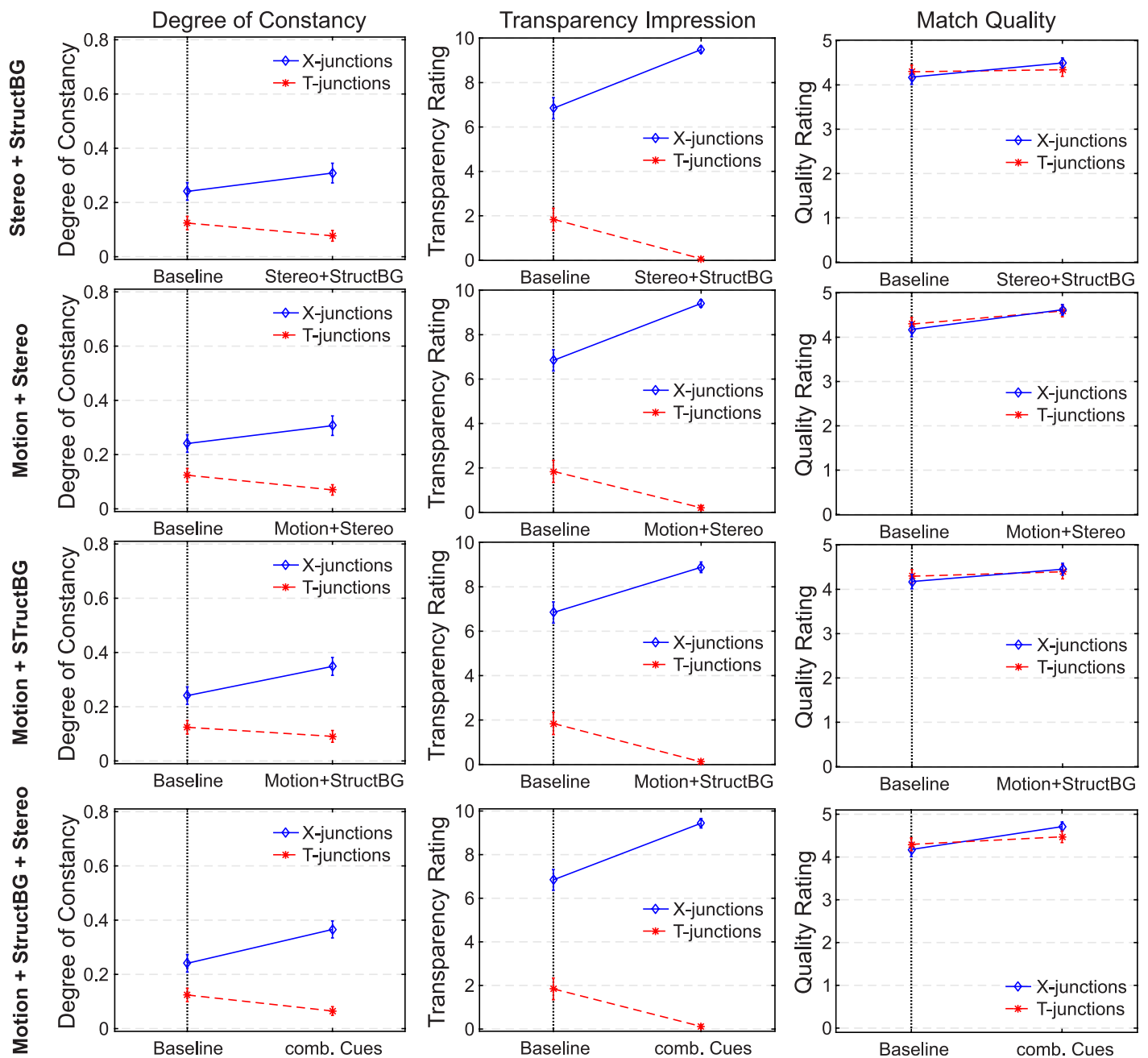


Figure A2. Results of Subexperiment 1, Part 1. The upper three rows correspond to the three two-cue-conditions in which two supporting cues were combined respectively: stereo and a structured background (StructBG), motion and stereo, and motion and a structured background. The columns correspond to the dependent variables: degree of constancy, transparency impression, and match quality, plotted separately for objects with X-junctions (blue open diamond, solid line) and T-junctions (red asterisks, dashed line). Error bars represent ± 2 SEM. The last row shows the combined cue condition for comparison.

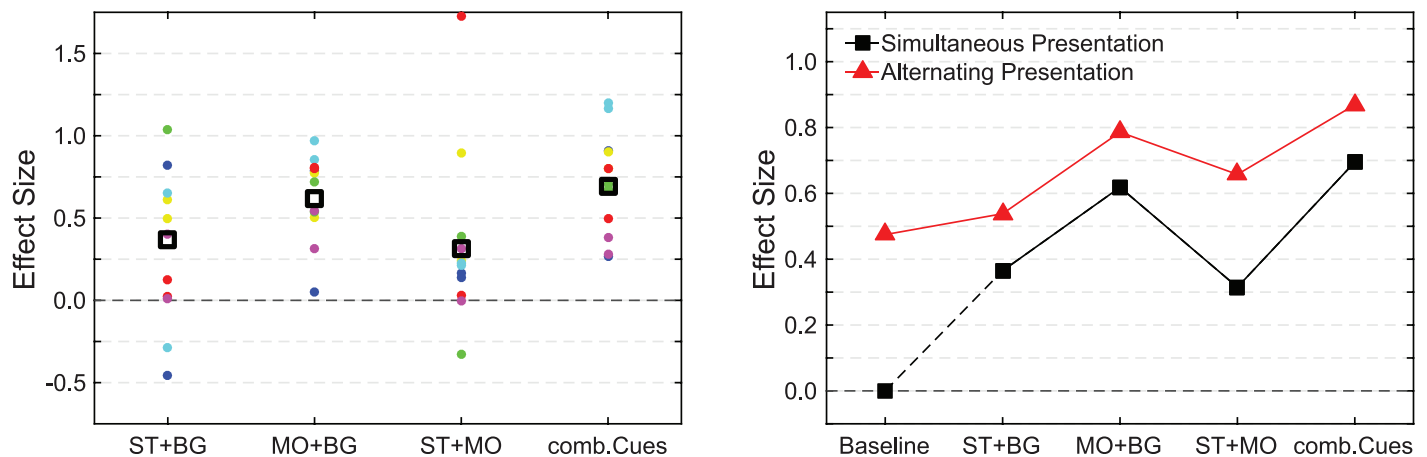


Figure A3. Effect sizes of the valid figural cue condition for each combination of two decomposition enhancing cues Stereo (ST), structured Background (BG), and Motion (MO). In the left panel, open squares indicate the effect size, Cohen's d for paired samples, of the each two-cue-combination, as well as the combined cue condition (ST + BG + MO). Each cue combination is compared pairwise to the baseline condition ($N = 108$). The dots show the effect sizes separately for all 12 filter-illumination-change combinations, colored according to the respective filter ($N = 9$). The right panel shows the effect sizes for the different cue conditions due to enhanced decomposition with simultaneous presentation (black squares) and the combination of enhanced decomposition and alternating presentation (red triangles).

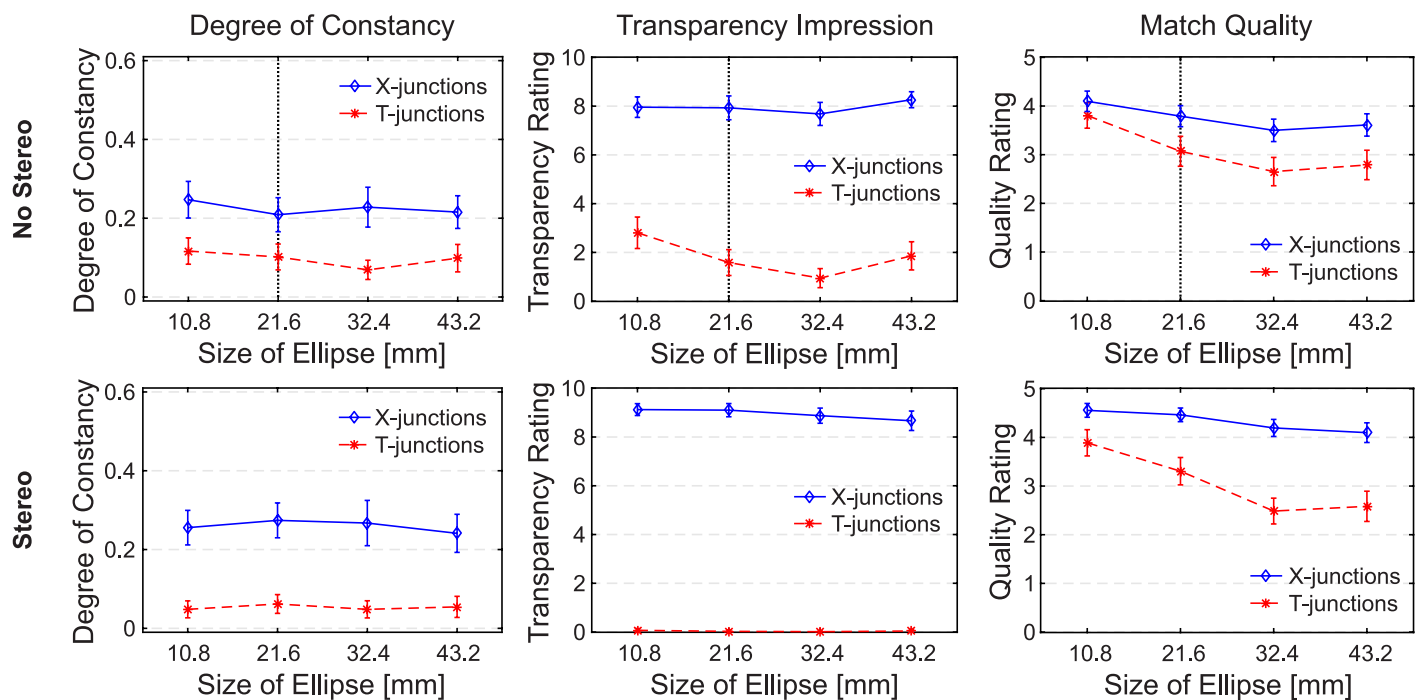


Figure A4. Data of the six subjects that processed the stimuli with different ellipse sizes with and without binocular disparity in Subexperiment 2, Part 1. Each data point corresponds to the mean of 72 measurements (6 subjects $\times$ 12 filter-illumination-combinations). The upper row shows the results without binocular disparity and the lower row with binocular disparity. The columns correspond to the dependent variables: degree of constancy, transparency impression, and match quality, plotted separately for objects with X-junctions (blue open diamond, solid line) and T-junctions (red asterisks, dashed line). Error bars represent ± 2 SEM.

Appendix E: Framework-dependent estimation of the filter parameter τ

The *framework-recognition hypothesis* states that the reason for the lower degrees of constancy with simultaneous presentation of standard and test is that the local illumination framework is ambiguous. In the extreme case the visual system perceives one *global illumination framework* instead of two different local illumination frameworks. The parameter of the standard filter $\hat{\tau}_{S,i}$ would then be calculated by

$$\hat{\tau}_{S,i} = \frac{\text{mean}(P_i)}{\text{mean}(A_i \cup B_i)}, i = L, M, S, \quad (11)$$

where all background colors from both halves of the screen are set in relation to the filtered colors. This estimate deviates from the original calculation according to Equation 9. In principle, all hybrid forms of illumination frameworks between the *local illumination framework* and the *global illumination framework* are possible. Therefore, separate filter parameters $\hat{\tau}_{S,i,\beta}$ are calculated for each possible mixing ratio of the background colors of standard $\text{mean}(A_i)$ and test $\text{mean}(B_i)$ as a function of β by

$$\hat{\tau}_{S,i,\beta} = \frac{\text{mean}(P_i)}{\text{mean}\left[\frac{A_i}{2} + \frac{(1-\beta)A_i}{2} + \frac{\beta B_i}{2}\right]}, i = L, M, S, \quad (12)$$

where $0 \leq \beta \leq 1$. The value $\beta = 1$ corresponds to a 100% *global illumination framework* and $\beta = 0$ corresponds to a 100% *local illumination framework*. The parameter of the test filter $\hat{\tau}_{T,i,\beta}$ are calculated in the same way, but with different weightings of the two

background halves:

$$\hat{\tau}_{T,i,\beta} = \frac{\text{mean}(Q_i)}{\text{mean}\left[\frac{B_i}{2} + \frac{(1-\beta)B_i}{2} + \frac{\beta A_i}{2}\right]}, i = L, M, S. \quad (13)$$

The colors $\text{mean}(P_i)$, $\text{mean}(A_i)$, and $\text{mean}(B_i)$ are determined from the fixed parts of the stimulus. The remaining filter color $\text{mean}(Q_i)$ is calculated for all resulting constancy predictions depending on β by

$$\begin{aligned} \text{mean}(Q_i, \beta) &= \frac{\text{mean}(P_i) \times \text{mean}[2B_i - \beta(B_i + A_i)]}{\text{mean}[2A_i - \beta(A_i + B_i)]}, \\ i &= L, M, S. \end{aligned} \quad (14)$$

For a 100% *local illumination framework* ($\beta = 0$) a constancy filter is obtained that is identical to the original prediction $\tau_{Standard}$ of the filter:

$$\begin{aligned} \text{mean}(Q_i, 0) &= \frac{\text{mean}(P_i) \times \text{mean}(B_i)}{\text{mean}(A_i)} \\ &= \tau_{Standard,i} \text{mean}(B_i), \\ i &= L, M, S. \end{aligned} \quad (15)$$

For a 100% *global illumination framework* ($\beta = 1$) a constancy filter is obtained that is identical to the proximal match:

$$\begin{aligned} \text{mean}(Q_i, 1) &= \frac{\text{mean}(P_i) \times \text{mean}(B_i + A_i)}{\text{mean}(A_i + B_i)} \\ &= \text{mean}(P_i), \\ i &= L, M, S. \end{aligned} \quad (16)$$